

M365 Administration Project

Day-to-day administration of a hybrid Microsoft 365 environment — mail flow, collaboration, and data governance

Exchange Online · Microsoft Defender · SharePoint Online
Microsoft Purview · Entra ID · Public DNS

SCOPE	1 prerequisite + 7 hands-on administration tasks
ENVIRONMENT	Microsoft 365 Business Premium, hybrid identity via Entra Connect
DURATION	~9 hours hands-on, single operator, live tenant
FINDINGS	9 root-cause findings · 7 corrections · 1 build-order revision
DATES	22–23 August 2026

Names are placeholders. The lab's real public domain, tenant and external test addresses have been replaced throughout with `contoso.beer`, `contoso.onmicrosoft.com` and `example.com` so the procedures transfer to a real environment. Screenshots retain the lab's own values.

At a glance

Objective	Move from <i>building</i> a hybrid environment to <i>administering</i> one — the work that generates actual tickets
Scope	1 prerequisite + 7 hands-on tasks across four admin portals not previously touched
Duration	Approximately 9 hours of hands-on work across two days
Environment	Microsoft 365 Business Premium, hybrid identity via Entra Connect, 2 domain controllers, 2 Windows 11 clients
Method	Single operator, live tenant, every failure traced to root cause before moving on
Findings	9 root-cause findings, 7 corrections to prior understanding, 1 revision to the build-order playbook
Deliverables	This document + a tagged field-manual integration module

Why this project exists

The previous project built a hybrid Active Directory and Microsoft 365 environment from bare metal, twice, and produced a build playbook. That work answered *how do you stand this up*.

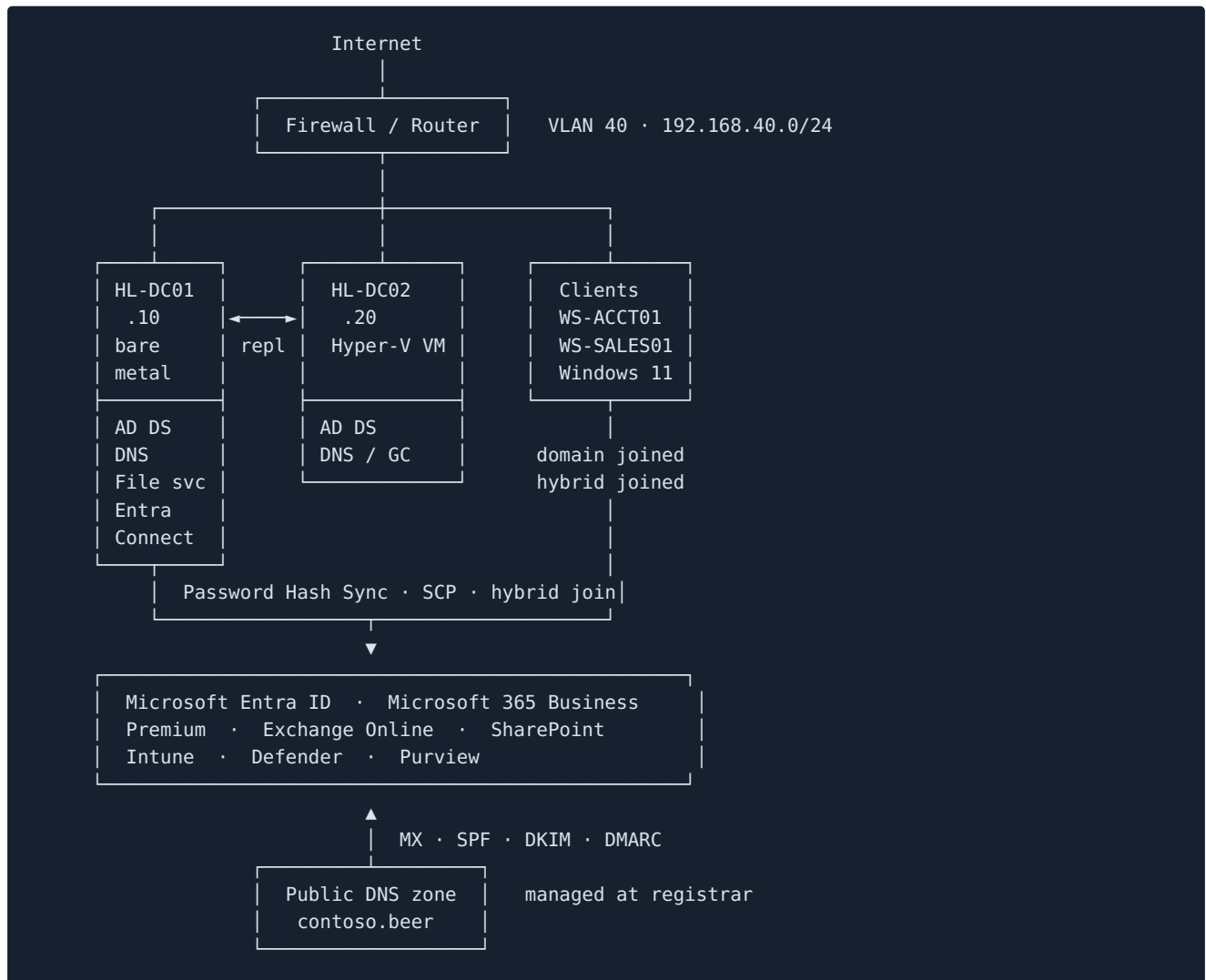
It did not answer *what do you do on Tuesday*.

At a small business, email is the number one ticket category. Shared mailboxes, distribution lists, "where did my message go," "why is this in quarantine," "can you give Sarah access to the sales inbox," "someone shared a file outside the company." None of that is build work. All of it is the job.

I had zero hands-on with any of it. So this project deliberately targeted the gap: **email, collaboration, and data governance**, in a live tenant, with real failures.

The output that matters is not the configuration. It is the set of findings — particularly the ones where a system reported success while doing nothing useful.

Environment



Namespace. Public domain `contoso.beer`; internal AD domain is the subdomain `ad.contoso.beer`. Public DNS is managed at the registrar, not delegated to Microsoft — a decision revisited in Task 0 and a deliberate one by the end of it.

Test population. `jsmith` (Sales) and `sjohnson` (Accounting), both synced from on-premises AD. `sysadmin@contoso.beer` as the daily-driver admin. A separate break-glass Global Administrator on the tenant's `.onmicrosoft.com` domain.

Task sequence and results

#	Task	Result	Root-cause findings
0	Configure the domain for Exchange Online — MX, autodiscover, SPF, DKIM, DMARC	Complete. SPF/DKIM/DMARC all pass	5
1	Run and read a message trace	Complete	1
2	Trip the spam filter, release from quarantine	Complete	1
3	Shared mailbox with Full Access, Send As, Send on Behalf	Complete	2
4	Distribution group vs mail-enabled security group	Complete	1
5	Add an alias, change the primary SMTP address	Complete	1 (+1 open)
6	Break SharePoint inheritance, share externally, locate the governing policy	Complete	2
7	Sensitivity label and DLP policy in Purview	Complete, 1 item deferred	2

Task 0 — Domain configuration for Exchange Online

Starting symptom, known before the project began: mail sent *from* @contoso.beer worked. Mail sent *to* it never arrived.

What the inventory found

The domain had been verified in Microsoft 365 months earlier — the ownership TXT record was published and the admin center accepted the domain. That was enough to set a UPN suffix and run Entra Connect, so the work had stopped there.

Domains

[+ Add domain](#) [Buy domain](#) [Refresh](#)

	Domain name ↑		Status	Choose columns
☐	homebiz.beer (Default)	:	ⓘ Incomplete setup	
☐	homebizbeer.onmicrosoft.com	:	✓ Healthy	

The tenant's own assessment: [contoso.beer \(Default\) – Incomplete setup](#).

Verification proves ownership and does nothing else. Outbound mail from Exchange Online requires none of the public DNS record set. Inbound requires all of it.

homebiz.beer		Switch to classic view ✕		
DNS Powered by Cloudflare				
Filter records...		+ Add record		
TYPE	HOST	ANSWER / VALUE	TTL	
ALIAS	homebiz.beer	pixie.porkbun.com	600	✎ ✖
CNAME	*.homebiz.beer	pixie.porkbun.com	600	✎ ✖
TXT	homebiz.beer	MS=ms27365663	3600	✎ ✖
Microsoft Entra Tenant				
TXT	_acme-challenge.homebiz.beer	0AqyZbtpZdGaFIF945nlvzg_hcAD XO1Suw wl-kTxn3o	600	✎ ✖
TXT	_acme-challenge.homebiz.beer	hMEzaY2EDqBWN7rzRQSZrmcYGeHZcsAR 1OYOmLx3QBE	600	✎ ✖

The zone before any work: an apex ALIAS to a parking page, a wildcard CNAME, the Microsoft verification TXT, and two stale ACME challenge records. No MX. No SPF. No DKIM. No DMARC.

Finding 1 — a wildcard CNAME made every diagnostic return a false positive

The inventory script was expected to return `DNS name does not exist` for the five missing records. It returned valid answers for all of them.

```
Administrator: Windows Powe x + v
PS C:\Users\evan> Resolve-DnsName -Name homebiz.beer -Type MX -Server 1.1.1.1

Name                               Type TTL  Section PrimaryServer      NameAdministrator      Serial
-----                               -
homebiz.beer                       SOA  1800  Authority curitiba.ns.porkbun.com dns.cloudflare.com      24
                                         12
                                         51
                                         51
                                         39

PS C:\Users\evan> Resolve-DnsName -Name autodiscover.homebiz.beer -Type CNAME -Server 1.1.1.1

Name                               Type  TTL  Section  NameHost
-----                               -
autodiscover.homebiz.beer          CNAME 600  Answer  pixie.porkbun.com

PS C:\Users\evan> Resolve-DnsName -Name homebiz.beer -Type TXT -Server 1.1.1.1 | Select-Object -ExpandProperty Strings
MS=ms27365663
PS C:\Users\evan> Resolve-DnsName -Name _dmarc.homebiz.beer -Type TXT -Server 1.1.1.1 | Select-Object -ExpandProperty Strings
Select-Object : Property "Strings" cannot be found.
At line:1 char:71
+ ... eer -Type TXT -Server 1.1.1.1 | Select-Object -ExpandProperty Strings
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (Microsoft.DnsCl...s.DnsRecord_PTR:PSObject) [Select-Object]
+ PSArgumentException
+ FullyQualifiedErrorId : ExpandPropertyNotFound,Microsoft.PowerShell.Commands.SelectObjectCommand

Select-Object : Property "Strings" cannot be found.
At line:1 char:71
+ ... eer -Type TXT -Server 1.1.1.1 | Select-Object -ExpandProperty Strings
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (Microsoft.DnsCl...s.DnsRecord_SOA:PSObject) [Select-Object]
+ PSArgumentException
+ FullyQualifiedErrorId : ExpandPropertyNotFound,Microsoft.PowerShell.Commands.SelectObjectCommand

PS C:\Users\evan> Resolve-DnsName -Name selector1._domainkey.homebiz.beer -Type CNAME -Server 1.1.1.1

Name                               Type  TTL  Section  NameHost
-----                               -
selector1._domainkey.homebiz.b     CNAME 600  Answer  pixie.porkbun.com
eer

PS C:\Users\evan> Resolve-DnsName -Name selector2._domainkey.homebiz.beer -Type CNAME -Server 1.1.1.1

Name                               Type  TTL  Section  NameHost
-----                               -
selector2._domainkey.homebiz.b     CNAME 600  Answer  pixie.porkbun.com
eer
```

Query	Answer received	Actual state
autodiscover CNAME	pixie.porkbun.com	Does not exist
selector1._domainkey CNAME	pixie.porkbun.com	Does not exist
selector2._domainkey CNAME	pixie.porkbun.com	Does not exist
_dmarc TXT	Select-Object : Property "Strings" cannot be found	Does not exist

The `*.contoso.beer` wildcard CNAME — the registrar's default parking record — answered for every subdomain without an explicit record.

The `Select-Object` error was not a syntax mistake. `Resolve-DnsName` returns a different .NET type per record type. The wildcard returned a CNAME, which PowerShell models as `DnsRecord_PTR` — it has a `NameHost` property, not `Strings`. **The error was the finding.**

The MX query behaved differently, and the difference is diagnostic:

NXDOMAIN — the name does not exist. **NODATA** — the name exists but has no record of that type. Proven by the zone SOA appearing in the *Authority* section with nothing in *Answer*.

Wildcards never match the zone apex, so the apex MX query returned honest NODATA while every subdomain lied.

Impact beyond the immediate task: `enterpriseregistration.contoso.beer` and `enterpriseenrollment.contoso.beer` — the discovery endpoints for device registration and Intune enrollment — were both resolving to a parking page. Nothing was broken, because hybrid join uses the Service Connection Point in AD and Intune enrollment came from Group Policy. It was a latent hazard that would never have been found.

Finding 2 — the bounce message was a textbook implicit-MX failure

Undelivered Mail Returned to Sender

MD

MAILER-DAEMON@proton.me
To: evan@evanireland.tech

email-1.3.eml (2.99 KB)
Outlook item

Reply

Reply All

Forward

...

Sat 22-Aug-26 11:52 AM

This is the mail system at Proton Mail.

Your email could not be delivered to one or more recipients.
See below for the delivery error and email.

The most common reasons for undelivered email are wrong recipient address or wrong configuration on the recipient mail server.

Learn more about delivery errors in our knowledge base: <https://proton.me/support/smtp-errors>

<sysadmin@homebiz.beer>: connect to homebiz.beer[207.207.210.229]:25:
Connection timed out

message/delivery-status

Reporting-MTA: dns; mail-4321.protonmail.ch
X-Postfix-Queue-ID: 4hQr9G410gzMrHMj
X-Postfix-Sender: rfc822; evan@evanireland.tech
Arrival-Date: Thu, 20 Aug 2026 17:32:18 +0000 (UTC)

Final-Recipient: rfc822; sysadmin@homebiz.beer
Original-Recipient: rfc822;sysadmin@homebiz.beer
Action: failed
Status: 4.4.1
Diagnostic-Code: X-Postfix; connect to homebiz.beer[207.207.210.229]:25:
Connection timed out

```
<sysadmin@contoso.beer>: connect to contoso.beer[203.0.113.10]:25:
Connection timed out
Status: 4.4.1
```

With no MX record, RFC 5321 says the sender falls back to the domain's address record — the **implicit MX rule**. The apex ALIAS resolved to the registrar's parking web server, which is not listening on port 25. The connection hung until timeout.

4.4.1 is a 4.x.x code — transient, not permanent. The original message arrived at Proton on 20 August at 17:32 UTC; the NDR was issued on 22 August at 11:52 UTC. Roughly 42 hours of retries before it gave up.

A **4.x.x** bounce means the mail is still coming if you fix the problem inside the retry window.

A **5.x.x** bounce means it is gone. That distinction changes what you tell a client on the phone.

Finding 3 — NS records in the zone are not a nameserver delegation

The first repair attempt chose Microsoft-managed DNS, which requires delegating the zone. The four Microsoft nameservers were entered into the **DNS record editor** as NS records.

Forty-five minutes of waiting produced nothing, because nothing had been delegated.

	Where it lives	How you change it
Delegation	The parent zone at the registry	The registrar's nameserver field, pushed via EPP
NS records in the zone	Your own zone	The DNS record editor

Microsoft's error message named the servers it was querying — the registrar's, not its own. The registrar's domain details page still listed the registrar's four nameservers.

NS records in the zone are meant to *mirror* the delegation. Pointing them at a provider the registry does not know about creates a **lame delegation**, and some resolvers will cache the child NS records in preference to the parent's and begin querying servers with no zone data.

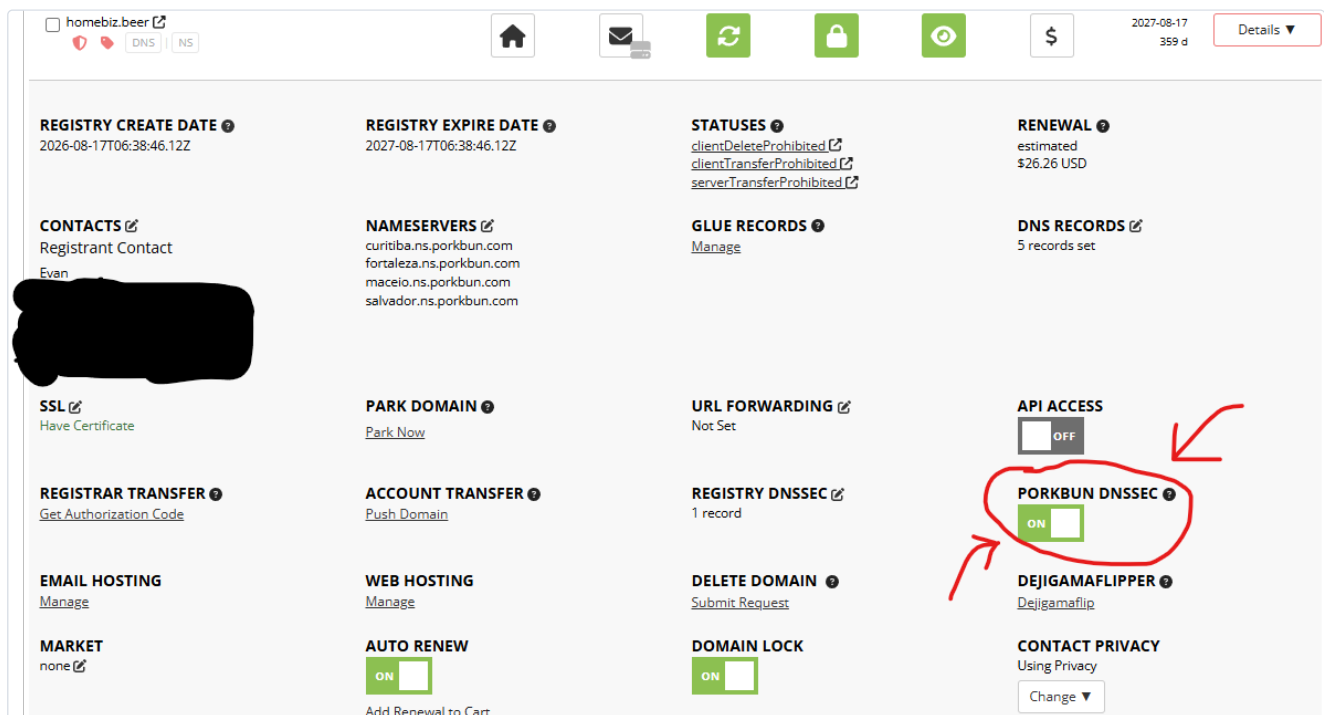
Finding 4 — DNSSEC, read the screen

ⓘ We've detected that homebiz.beer has Delegation of Signing (DS) DNS records, used for Domain Name System Security Extensions (DNSSEC), configured at your domain's registrar or previous DNS hosting provider. DNSSEC is not supported by Microsoft 365 DNS hosting. To continue setting up Microsoft 365 to manage your DNS records, go to your domain registrar or DNS hosting provider and disable DNSSEC or delete the DS records. If you recently deleted the DS records, please allow up to 48 hours for the changes to be reflected here.

We queried these nameservers for the records: **fortaleza.ns.porkbun.com**, **salvador.ns.porkbun.com**

"We've detected that contoso.beer has Delegation of Signing (DS) DNS records... DNSSEC is not supported by Microsoft 365 DNS hosting."

The banner had been on screen the entire time.



From the project log, verbatim:

"So I just noticed that a little warning message in the M365 Admin Center clearly says... So uh...yeah I turned it off in Porkbun. My bad! Read the screens in front of you!"

Then it produced a second, more interesting failure. DNSSEC removal has a mandatory order: remove the DS record from the parent registry, wait out its TTL, *then* stop signing. Reversing it means the parent still advertises "this zone is signed — reject unsigned answers," while the zone no longer signs. Every validating resolver returns **SERVFAIL** and the domain goes dark for a large share of the internet.

The verification caught exactly this, mid-transition:

Resolver	Result
Cloudflare 1.1.1.1	SERVFAIL
Google 8.8.8.8	Clean SOA
Parent zone DS query	NODATA — .beer holds no DS record

SERVFAIL from one public resolver with a clean answer from another is the signature of DNSSEC validation, essentially every time. A genuinely down nameserver fails everywhere. Two resolvers disagreeing means one is validating against something the other is not — usually a stale cached DS.

homebiz.beer

Updated: 2026-08-23 01:22:27 UTC (less than a minute ago) [Update now](#)

« Previous analysis | Next analysis »

2026-08-23 Go

DNSSEC Responses Servers Analyze

[Tweet](#)

— DNSSEC options ([show](#)) —

Notices

RRset status

☒ Insecure (3)

☒ Secure (2)

DNSKEY/DS/NSEC status

☒ Secure (7)

Delegation status

☒ Insecure (1)

☒ Secure (1)

DNSKEY legend

[Full legend](#)

☒ SEP bit set

☒ Revoke bit set

☒ Trust anchor

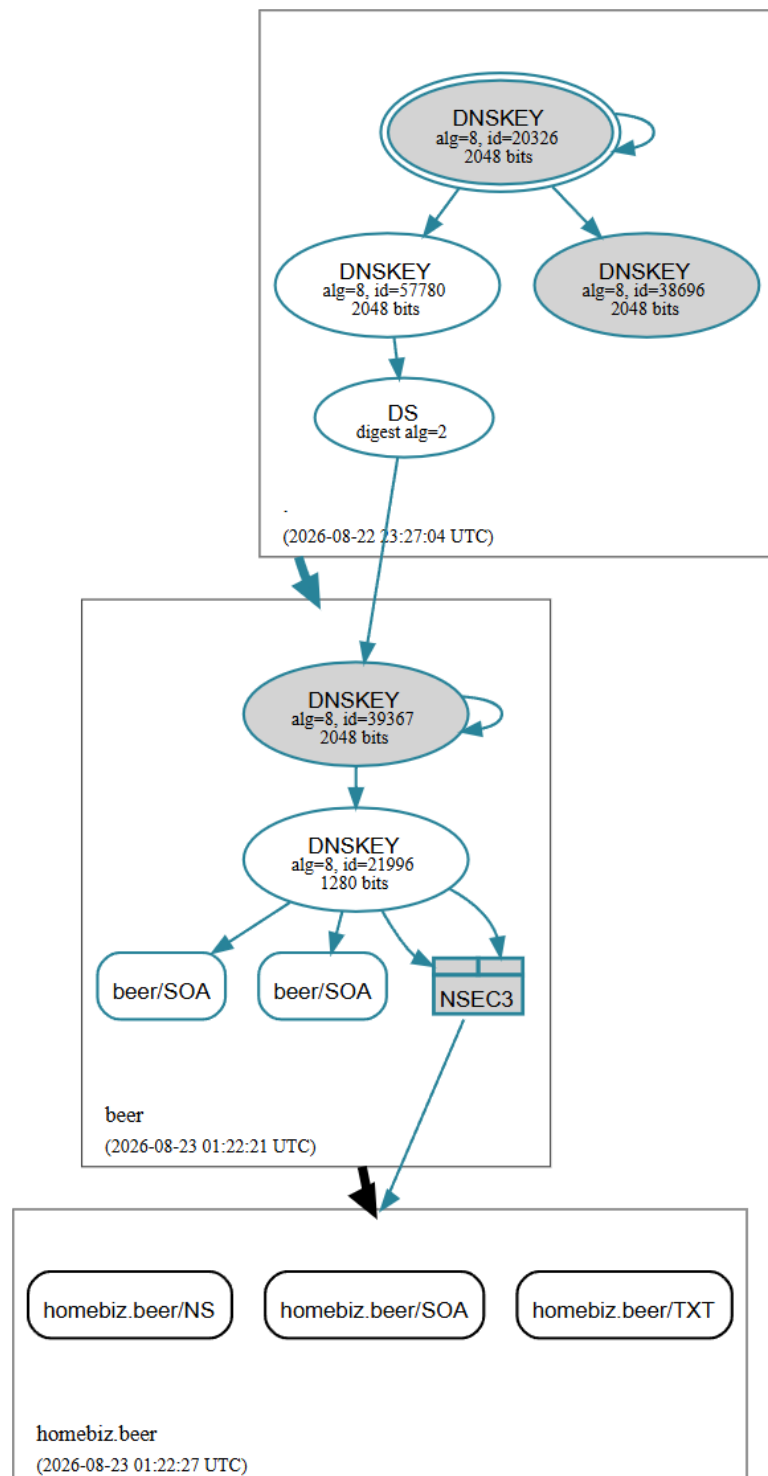
See also

[DNSSEC Debugger](#) by Verisign Labs.

DNSSEC Authentication Chain

Download: [png](#) | [svg](#)

Mouse over and click elements in the graph below to see more detail.



dnsviz.net renders the full delegation chain from root down — the right tool for any real DNSSEC ticket, rather than inferring from resolver behavior.

The strategic decision: abandon Microsoft-managed DNS and keep the zone at the registrar. Four reasons, in order of weight:

1. **It is the skill the job actually requires.** Real clients keep DNS wherever their web developer put it a decade ago. Delegating to Microsoft means you can no longer create a record for anything that is not Microsoft.
2. **DNSSEC stays available.** Microsoft-hosted DNS does not support it.
3. **Speed.** Three records at a 600-second TTL versus a registry delegation plus a documented 48-hour DS-removal window.
4. It had already been done correctly once, on another domain, in minutes.

Finding 5 — DKIM CNAME values cannot be constructed

The record format changed in May 2025. New domains receive a selector target of the form:

```
selector1-contoso-beer._domainkey.contosobeer.p-v1.dkim.mail.microsoft
```

The character before `-v1` is a **dynamic partition character** assigned by Microsoft. It is not configurable and not predictable, and the old and new formats cannot coexist for the same selector. **The values must be read from the portal or from `Get-DkimSigningConfig`.**

homebiz.beer				Switch to classic view X	
DNS Powered by Cloudflare					
Filter records...				+ Add record	
TYPE	HOST	ANSWER / VALUE	TTL		
CNAME	autodiscover.homebiz.beer	autodiscover.outlook.com	3600		
CNAME	enterpriseenrollment.homebiz.beer	enterpriseenrollment-s.manage.microsoft.com	3600		
CNAME	enterpriseregistration.homebiz.beer	enterpriseregistration.windows.net	3600		
CNAME	selector1._domainkey.homebiz.beer	selector1-homebiz-beer._domainkey.homebizbeer.p-v1.dkim.mail.microsoft	3600		
CNAME	selector2._domainkey.homebiz.beer	selector2-homebiz-beer._domainkey.homebizbeer.p-v1.dkim.mail.microsoft	3600		
MX	homebiz.beer	homebiz-beer.mail.protection.outlook.com prio: 0	3600		
TXT	homebiz.beer	MS=ms27365663	3600		
Microsoft Entra Tenant					
TXT	homebiz.beer	v=spf1 include:spf.protection.outlook.com -all	3600		
TXT	_dmarc.homebiz.beer	v=DMARC1; p=none; rua=mailto:sysadmin@homebiz.beer	600		

Final zone: MX, autodiscover, SPF, both DKIM selectors, both Intune discovery records, the verification TXT. Wildcard, apex ALIAS, misplaced NS records, and stale ACME records all removed.

Result

Email authentication settings

ARC DKIM

DomainKeys Identified Mail (DKIM)

Domain Keys Identified Mail (DKIM) is an authentication process that can help protect both senders and recipients from forged and phishing email. Add DKIM signatures to your domains so recipients know that email messages actually came from users in your organization and weren't modified after they were sent. [Learn more about DKIM](#)

↓ Export ↻ Refresh 3 items 10 Items per page 🔍 Search ⌵

Name ▾	Accepted domain ▾	Domain Type ▾	Status ▾	Toggle ▾
☐ homebiz.beer	homebiz.beer	Authoritative	Valid	☒ Enabled
☐ homebizbeer.mail.onmicrosoft.com	homebizbeer.mail.onmicrosoft.com	Authoritative	NoDKIMKeys	☐ Disabled
☐ homebizbeer.onmicrosoft.com (default signing domain)	homebizbeer.onmicrosoft.com	Authoritative	NoDKIMKeys	☐ Disabled

DKIM signing enabled and valid for the custom domain. Note the two `onmicrosoft.com` rows below it — one is the tenant initial domain, one is the hybrid mail routing domain used for coexistence.

```
Administrator: Windows PowerShell
PS C:\Users\evan> Resolve-DnsName -Name autodiscover.homebiz.beer -Type CNAME

Name                Type  TTL  Section  NameHost
----                -
autodiscover.homebiz.beer  CNAME 3548 Answer  autodiscover.outlook.com

PS C:\Users\evan> Resolve-DnsName -Name homebiz.beer -Type MX

Name                Type  TTL  Section  NameExchange  Preference
----                -
homebiz.beer        MX    3549 Answer  homebiz-beer.mail.protection.outlook.com  0

Name                : homebiz-beer.mail.protection.outlook.com
QueryType           : A
TTL                 : 10
Section             : Additional
IP4Address          : 52.101.11.3

Name                : homebiz-beer.mail.protection.outlook.com
QueryType           : A
TTL                 : 10
Section             : Additional
IP4Address          : 52.101.8.34

Name                : homebiz-beer.mail.protection.outlook.com
QueryType           : A
TTL                 : 10
Section             : Additional
IP4Address          : 52.101.40.2

Name                : homebiz-beer.mail.protection.outlook.com
QueryType           : A
TTL                 : 10
Section             : Additional
IP4Address          : 52.101.50.7
```

Resolution verified through the real client path — workstation → domain controller → forwarder. The four `52.101.x.x` addresses carry a 10-second TTL because Microsoft rotates inbound protection endpoints constantly. This is why the MX value is a hostname and never an IP.

From Evan Ireland <evan@evanireland.tech> ☆ 19:50
To Administrator ▼

Testing

Best,

Evan Ireland
Email: evan@evanireland.tech
Website: evanireland.tech

Original Message

Message ID	<BY1P220MB19310C1207752F1FDA33EAC0ACA12@BY1P220MB1931.NAMP220.PROD.OUTLOOK.COM>
Created at:	Sat, Aug 22, 2026 at 8:08 PM (Delivered after 3 seconds)
From:	Administrator <sysadmin@homebiz.beer>
To:	
Subject:	DMARC Test
SPF:	PASS with IP 2a01:111:f403:c107:0:0:0:3 Learn more
DKIM:	'PASS' with domain homebiz.beer Learn more
DMARC:	'PASS' Learn more

SPF, DKIM, and DMARC all pass. Two hours earlier this domain was timing out against a parking page.

Domains

[+ Add domain](#) [📁 Buy domain](#) [🔄 Refresh](#)

	Domain name ↑		Status
☐	homebiz.beer (Default)	:	✓ Healthy
☐	homebizbeer.onmicrosoft.com	:	✓ Healthy

One detail worth catching: across two outbound tests sixteen minutes apart, the SPF-authorized source IP changed from `2a01:111:f403:c100::f` to `2a01:111:f403:c107::3`. Microsoft rotates outbound sending addresses continuously. **This is exactly why the SPF record uses `include:` and never hardcoded `ip4:` entries.** Anyone who pins Microsoft IPs into SPF has signed up for a silent outage on the day those IPs change.

DMARC deployment — the template that came out of this

```
Host: _dmarc
Type: TXT
Value: v=DMARC1; p=none; rua=mailto:dmarc@contoso.com
TTL: 600
```

`p=none` is **monitoring only. It protects nothing.** Its entire value is that aggregate reports tell you who is sending as the domain before you start blocking anything.

Stage	Record	Duration	Exit criteria
Monitor	<code>p=none; rua=...</code>	2-4 weeks	Every legitimate sender identified in reports
Authorize	unchanged	as needed	Each sender fixed in SPF or DKIM
Partial	<code>p=quarantine; pct=10 → 25 → 50</code>	1-2 weeks per step	No legitimate mail quarantined
Enforce	<code>p=quarantine; pct=100</code> , then <code>p=reject</code>	—	—

Skipping to `p=reject` is how you discover on a Monday that the client's invoicing system was sending as their domain from a server nobody documented.

Build-order revision

Domain DNS configuration belongs in Phase 0, not deferred.

Verifying a domain is sufficient to set a UPN suffix and run Entra Connect, so it is tempting to stop there and continue with identity work. That defers work that must be done anyway and does it later under worse conditions — with users already synced and expecting a working mailbox. Complete the full record set at the same sitting as domain verification. **Check for DNSSEC before choosing a DNS hosting model**, since the incompatibility only surfaces after the migration has started.

Task 1 — Message trace

The tool you will open most often at an MSP, because "where's my email" is the most common ticket there is.

Message trace search results				
The new Message Trace is now available and is turned on by default. Learn more				
Export results Edit message trace Refresh		4 items Customize columns Filter Search		
Date (UTC-07:00) ↓	Sender	Recipient	Subject	Status
8/22/2026, 8:22 PM	sysadmin@homebiz.beer	sysadmin@homebiz.beer	Internal Email Test	Delivered
8/22/2026, 8:08 PM	sysadmin@homebiz.beer	[REDACTED]	DMARC Test	Delivered
8/22/2026, 7:52 PM	sysadmin@homebiz.beer	[REDACTED]	Test (Outbound)	Delivered
8/20/2026, 10:16 PM	sysadmin@homebiz.beer	evan@evanireland.tech	Test	Delivered

The hard boundary

Message trace shows what Exchange Online Protection and Exchange Online did with a message **while it was inside the service**. It stops at the mailbox door.

It cannot see inbox rules, Sweep rules, Focused vs Other, a user dragging a message somewhere, or a client-side rule in Outlook desktop. It cannot see what an external server did after accepting the handoff.

The most common misread in the field: the ticket says "it never arrived," the trace says **Delivered**. The trace is not lying. The message is in the mailbox and something after delivery moved it — nine times out of ten an inbox rule the user forgot they created. Pivot to the mailbox, not to mail flow.

Reading the events

An internal self-addressed message and an inbound internet message produce visibly different chains:

Internal Email Test

 Copy report text  Prepare and email extended report

 View message in Explorer  Go Hunt for this message

Sender

sysadmin@homebiz.beer

Recipient

sysadmin@homebiz.beer

Received

Processed

Delivered

Status

The message was delivered to the recipient's Inbox folder.

Delivery time: 08/23/2026 02:22:45 (UTC)

More Information

If the recipient can't find the message in their Inbox folder, it might have been deleted or moved to another folder (such as Junk Email) either manually or automatically based on an Inbox rule or Sweep rule the recipient set up. Ask them to search for the message across all folders in their mailbox.

Tip: If the recipient still can't find the message in Outlook, they might be having connectivity issues. Ask them to try restarting Outlook or use [Outlook on the web](#) to check for the message. To see detailed steps for fixing Outlook, see [Fix Outlook connection problems in Office 365](#).

Message events

>	Date (UTC-07:00)	Event	Detail
>	8/22/2026, 8:22 PM	Receive	Message received by...
>	8/22/2026, 8:22 PM	Submit	The message was su...
>	8/22/2026, 8:22 PM	Deliver	The message was su...

More information

Message ID

<BY1P220MB1931611CD6329DD5A364F8F3ACA12@BY1P220MB1931.NAMP220.PROD.OUTLOOK.COM>

MessageTrace ID

7992d93a-2472-4d34-f2c4-08df00bd698a

Message size

34.85 KB

From IP

M365 Administration Project - HomeBiz Lab Phase 2

To IP

Internal self-send: Receive → Submit → Deliver, with both identifier fields visible at the bottom.

Trace	Events
Internal self-send	Receive → Submit → Deliver
Inbound from Proton	Receive → Deliver
Outbound to external	Receive → Submit → TRANSFER → Send external

Finding — the Submit event is an origin test. Submit is where a message is handed into transport *by a mailbox inside the organization*. An inbound internet message has no local mailbox submitting it.

Submit present = originated inside the tenant. Submit absent = arrived from the internet. A message claiming to be from an internal sender with no Submit event did not originate from an internal mailbox. Directly useful for "is this spoofed, or did it really come from our CEO?"

Deliver vs Send external is the other distinction to keep. Deliver means it is in a mailbox you administer. Send external means you successfully handed it to somebody else's server and your visibility ends there.

Two IDs, two purposes

Field	Assigned by	Use
Message-ID	The originating mail system. Never changes.	The highest-precision search key available. Have the user open the original and copy the header.
MessageTrace ID	A GUID internal to Exchange Online's trace database.	Hand to Microsoft Support when escalating.

A message originating at Proton kept its [@example.com](#) Message-ID all the way through Exchange delivery. A message originating in Exchange Online carried the Microsoft format.

Retention and licensing

Age	Behavior
Under 7 days	Interactive results in the browser
7-90 days	Downloadable CSV only, up to an hour to generate
Over 90 days	Gone. Not archived, not recoverable

Licensing flag: *View message in Explorer* requires E5-tier Threat Intelligence. *Go Hunt* requires Defender for Office 365 **Plan 2**. Business Premium includes **Plan 1**. *Copy report text* and *Prepare and email extended report* work fine.

Time zone trap: the event list header labels the zone by its **standard** offset while rendering times in **daylight** time. A header reading `Date (UTC-07:00)` beside a delivery time of `01:50 UTC` displays `7:50 PM`, not `6:50 PM`. **Correlate in UTC.** Same class of failure as Kerberos rejecting a domain join over clock skew — time handling produces failures that present as something else entirely.

Task 2 — Spam filtering and quarantine



Anti-spam inbound policy (Default)

Always on | Priority Lowest

SPF record: hard fail

Off

Conditional Sender ID filtering: hard fail

Off

Backscatter

Off

Test mode action

None

International spam - languages

Off

International spam - regions

Off

[Edit spam threshold and properties](#)

Actions

Spam message action

Move message to Junk Email folder

High confidence spam message action

Move message to Junk Email folder

Phishing message action

Move message to Junk Email folder

High confidence phishing message action

Quarantine message

Apply the following quarantine policy:

AdminOnlyAccessPolicy

Bulk message action

Move message to Junk Email folder

Bulk moves enabled

NotSet

Intra-Organizational messages to take action on

Default

Default policy state recorded before any modification — the baseline you need in order to put it back.

Junk and quarantine are different places

	Location	User can retrieve it?
Junk Email	Inside the mailbox	Yes, themselves
Quarantine	Held by the service — never reaches the mailbox	Depends on the quarantine policy

The distinction surfaced as two different message trace statuses on the same sender, seven minutes apart, with one variable changed between them:

The screenshot shows the Microsoft Defender Quarantine interface. On the left, a sidebar contains navigation links: Home, Incidents, Advanced hunting, and Show navigation. The main area is titled 'Quarantine' and shows a list of items. The selected item, 'Quarantine Test 2', is displayed in a detailed view on the right. This view includes a header with '1 Links' and actions like 'Open email entity', 'Take action', and 'Release email'. Below this, a 'Quarantine details' section provides information about the message, including its received time, expiration, subject, policy type, recipient, and quarantine reason.

Quarantine details		
Received	Aug 22, 2026 9:01:20 PM	Expires
Subject	Quarantine Test 2	Quarantine reason
Policy type	Anti-spam policy	Policy name
Recipient	sysadmin@homebiz.beer	Not yet released to
Sender address override reason	None	Released by

9:01 PM	Quarantine Test 2	Quarantined
8:54 PM	Quarantine Test	FilteredAsSpam

Finding — two policy objects, not one

This is the distinction juniors collapse, and it changes what you tell a user on the phone.

Object	Answers
Anti-spam policy	<i>Should this message be quarantined?</i>
Quarantine policy	<i>Once quarantined, what may the end user do about it?</i>

Message trace search results				
ⓘ The new Message Trace is now available and is turned on by default. Learn more ×				
↓ Export results ✎ Edit message trace ↻ Refresh 4 items ⚙ Customize columns ⏮ Filter Search ≡				
Date (UTC-07:00) ↓	Sender	Recipient	Subject	Status
8/22/2026, 9:01 PM	evan@evanireland.tech	sysadmin@homebiz.beer	Quarantine Test 2	Quarantined
8/22/2026, 8:54 PM	evan@evanireland.tech	sysadmin@homebiz.beer	Quarantine Test	FilteredAsSpam
8/22/2026, 8:22 PM	sysadmin@homebiz.beer	sysadmin@homebiz.beer	Internal Email Test	Delivered
8/22/2026, 7:50 PM	evan@evanireland.tech	sysadmin@homebiz.beer	Test (Inbound)	Delivered

Every quarantining action carries its own quarantine policy assignment:

Policy	User can
<code>AdminOnlyAccessPolicy</code>	Nothing. The message does not appear in their quarantine view at all.
<code>DefaultFullAccessPolicy</code>	View and release their own mail
<code>DefaultFullAccessWithNotificationPolicy</code>	Same, plus digest notifications

The default assignment for high-confidence phishing is `AdminOnlyAccessPolicy` — deliberately, so users cannot self-release the thing that encrypts the file server.

The correct answer to "can I get it back myself?" is "let me check — it depends how it was classified." Sending a user to a page that will show them nothing is a wasted callback.

Retention was 15 days, matching `Retain spam in quarantine for this many days` on the policy.
Configurable — read it off the policy rather than quoting a number.

Testing note: GTUBE, the widely cited spam-filter test string, is not documented as supported by Exchange Online Protection and did not trip it. A blocked-sender entry or a mail flow rule stamping SCL 9 are the deterministic methods.

Lab vs production: the Default anti-spam policy is always lowest priority and applies to everyone. In production you create a scoped policy at higher priority targeting a test user and delete it when finished — you do not edit Default.

Task 3 — Shared mailbox and delegation



Sales

sales@homebiz.beer



Basic information

Name

Sales

[Edit](#)

Email addresses

Primary

sales@homebiz.beer

Aliases

sales@homebizbeer.onmicrosoft.com

[Edit](#)

Email forwarding

None

[Edit](#)

Automatic replies

Off

[Edit](#)

Sent items

Not copied to mailbox

[Edit](#)

Email apps

All email apps allowed

[Edit](#)

Members

Administrator

John Smith

Sarah Johnson

[Edit](#)

Manage mailbox permissions

[Read and manage permissions \(3\)](#)

[Send as permissions \(2\)](#)

[Send on behalf of permissions \(1\)](#)

Show in global address list

Yes

[Edit](#)

Exchange settings

[Edit](#)

What a shared mailbox is

A real mailbox with a real Entra user object behind it, and **that account is blocked from sign-in**. Not a distribution group, not an alias.

Licensing: no license required under 50 GB. `info@`, `support@`, `billing@` cost nothing — one of the most useful cost facts in SMB work.

Security note that gets skipped: the hidden account is a standing target. The sign-in block is the control. Attackers specifically hunt for shared mailbox accounts where someone enabled sign-in "to make it easier."

Three permissions, three storage locations

Permission	Effect	Retrieved with
Full Access	Open, read, organize, delete. No right to send.	<code>Get-MailboxPermission</code>
Send As	Mail appears to come from the mailbox, full stop	<code>Get-RecipientPermission</code>
Send on Behalf	Adds a <code>Sender:</code> header naming the delegate	A property on the mailbox object

```
Administrator: Windows Power Shell
PS C:\Users\evan> Get-RecipientPermission -Identity sales@homebiz.beer

Identity      Trustee      AccessControlType AccessRights Inherited
-----
Sales20260823033230 NT AUTHORITY\SELF Allow {SendAs} False
Sales20260823033230 jsmith@homebiz.beer Allow {SendAs} False
Sales20260823033230 sjohnson@homebiz.beer Allow {SendAs} False

PS C:\Users\evan> Get-Mailbox -Identity sales@homebiz.beer | Select-Object -ExpandProperty GrantSendSendOnBehalfTo
Select-Object : Property "GrantSendSendOnBehalfTo" cannot be found.
At line:1 char:44
+ ... @homebiz.beer | Select-Object -ExpandProperty GrantSendSendOnBehalfTo
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (C@{AcceptMessage...ityDescriptor}:PSObject) [Select-Object], PSArgumentExcepti
on
+ FullyQualifiedErrorId : ExpandPropertyNotFound,Microsoft.PowerShell.Commands.SelectObjectCommand

PS C:\Users\evan> Get-Mailbox -Identity sales@homebiz.beer | Select-Object -ExpandProperty GrantSendOnBehalfTo
c2a950b5-319c-4676-aa27-8280e89cda85
PS C:\Users\evan> Set-Mailbox -Identity sales@homebiz.beer -MessageCopyForSentAsEnabled $true -MessageCopyForSendOnBehalfEnabled
$true
PS C:\Users\evan> Get-Mailbox -Identity sales@homebiz.beer | Format-List Name,MessageCopyForSentAsEnabled,MessageCopyForSendOnBehalfEnabled

Name                : Sales20260823033230
MessageCopyForSentAsEnabled : True
MessageCopyForSendOnBehalfEnabled : True

PS C:\Users\evan> Disconnect-ExchangeOnline

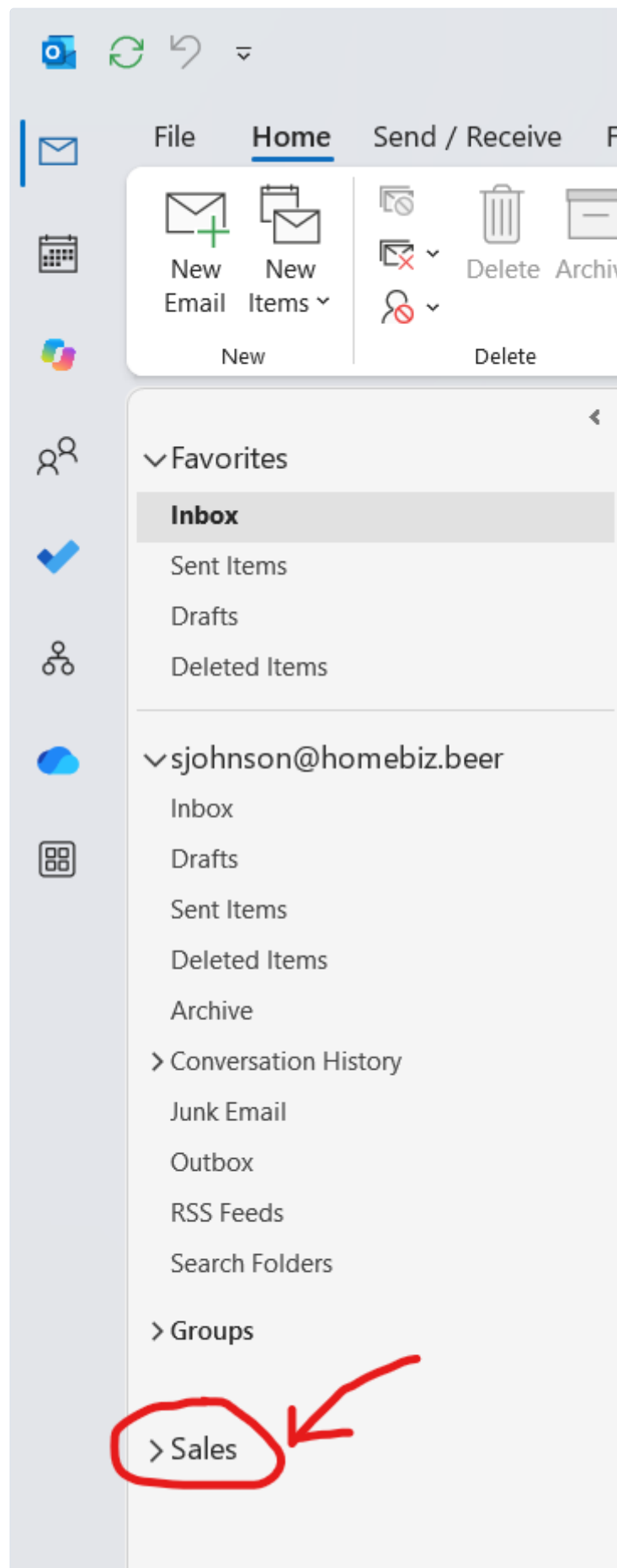
Running this cmdlet will clear the specified REST connections along with the connection for inbuilt cmdlets and all the active
RPS sessions created using Connect-ExchangeOnline or Connect-IPSSession.
Press(Y/y/A/a) if you want to continue.
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): A
PS C:\Users\evan>
```

Three permissions, three unrelated storage locations, three different retrieval methods. Send As is a directory-level right on the recipient object, not an ACL on the mailbox

store — which is why it needs a different cmdlet entirely. The GUI presents them as one tidy list, which is precisely why people miss one when auditing an inherited tenant.

If an account holds both Send As and Send on Behalf, Send As wins and Send on Behalf never fires.

Automapping worked as documented — Full Access granted directly to a user caused the mailbox to appear in Outlook desktop without configuration. It does not fire for permissions granted through a group.



The Sent Items problem

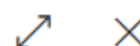
Default behavior sends a delegate's sent copy to **their** Sent Items only. Nobody else on the team sees it — the classic "we can't tell who already replied, so we keep double-answering" ticket.

```
Set-Mailbox -Identity sales@contoso.beer `
-MessageCopyForSentAsEnabled $true `
-MessageCopyForSendOnBehalfEnabled $true
```

Two separate switches. Setting one does nothing for the other. **Not retroactive.**

Finding — Send As erases the submitter from the audit trail

Original Message	
Message ID	<DM8P220MB06727420057B223D0E1724D2B8A12@DM8P220MB0672.NAMP220.PROD.OUTLOOK.COM>
Created at:	Sat, Aug 22, 2026 at 10:00 PM (Delivered after 2 seconds)
From:	Sales <sales@homebiz.beer>
To:	[REDACTED]
Subject:	Sales Test
SPF:	PASS with IP 2a01:111:f403:c10c:0:0:0:1 Learn more
DKIM:	'PASS' with domain homebiz.beer Learn more
DMARC:	'PASS' Learn more



Sales Test

Copy report text Prepare and email extended report

View message in Explorer Go Hunt for this message

Sender

sales@homebiz.beer

Recipient



Received

Processed

Sent

Status

Office 365 successfully sent the message to the following external address:

External address



More Information

If the recipient can't find the message in their Inbox folder, one of several things could have happened:

- The message may have been deleted or moved to another folder (for example, the Junk Email folder).
- The recipient might have created a forwarding rule that forwarded the message to a different email address.
- There might be something wrong with the recipient's email system.

Until the problem is resolved, ask the sender to try contacting the recipient another way. For instance, send the message to a different email address.

Message events



>	Date (UTC-07:00)	Event	Detail
>	8/22/2026, 10:00 PM	Receive	Message received...
>	8/22/2026, 10:00 PM	Submit	The message was ...
>	8/22/2026, 10:00 PM	TRANSFER	
>	8/22/2026, 10:00 PM	Send external	Message sent to ...

Sarah sent from the shared mailbox using **Send As**. Both the recipient's view and the message trace show `sales@contoso.beer`. **Sarah's name appears nowhere.**

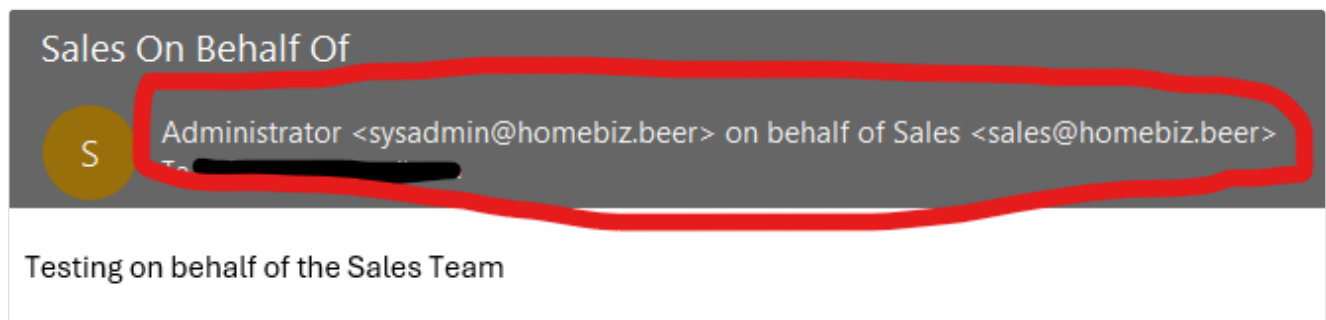
Send As does not add a sender — it **replaces** one. If three people hold Send As on `sales@` and one emails a client something they shouldn't have, **message trace cannot tell you which human it was.**

Recovering that attribution requires **mailbox auditing** — Exchange logs a `SendAs` record naming the delegate, searchable in the unified audit log. **Message trace and audit logs answer different questions.**

The same project produced the contrast four hours later, by accident. A Task 7 message sent from the same shared mailbox by an account holding **Send on Behalf** showed `sales@contoso.beer` at the recipient and `sysadmin@contoso.beer` in the trace. Same mailbox, opposite behavior, determined entirely by delegation type.

Finding — Send on Behalf attribution depends on the recipient's client

Per RFC 5322, Send on Behalf produces two address headers: `From:` naming the author (the mailbox) and `Sender:` naming the submitter (the delegate).



Outlook desktop parses `Sender:` and renders "Administrator on behalf of Sales."

Original Message	
Message ID	<BY1P220MB1931FAD4CA9A438E05A62AD9ACA12@BY1P220MB1931.NAMP220.PROD.OUTLOOK.COM>
Created at:	Sat, Aug 22, 2026 at 10:23 PM (Delivered after 8 seconds)
From:	Sales <sales@homebiz.beer>
To:	[REDACTED]
Subject:	Sales On Behalf Of
SPF:	PASS with IP 2a01:111:f403:c110:0:0:0:1 Learn more
DKIM:	'PASS' with domain homebiz.beer Learn more
DMARC:	'PASS' Learn more

Gmail's Show original summary table lists Message ID, Created at, From, To, Subject, and authentication results. **Sender:** is present only in the raw headers below — not in the summary a user would look at.

Same message. Same headers. Two clients, two different displays.

Send on Behalf preserves attribution in the message. It does not guarantee the recipient will ever see it. Where accountability actually matters — compliance, legal, regulated communications — the enforceable record is the mailbox audit log, not the recipient's rendering.

That is a materially different conclusion from the intended lesson. Send on Behalf is frequently recommended *for* accountability; this demonstrates the guarantee is weaker than advertised.

DMARC is unaffected — it aligns on the **From:** domain only. **Sender:** plays no part in it.

Task 4 — Distribution group vs mail-enabled security group

A distribution group is a mailing list. A mail-enabled security group is a mailing list that is also a security principal — an object that can hold permissions, be licensed, and be targeted by Conditional Access.

```
Administrator: Windows PowerShell
PS C:\Users\evan> Get-DistributionGroup | Format-List Name,RecipientTypeDetails,PrimarySmtpAddress,IsDirSynced

Name                : Announcements20260823152758
RecipientTypeDetails : MailUniversalDistributionGroup
PrimarySmtpAddress   : announcements@homebiz.beer
IsDirSynced          : False

Name                : Sales Team20260823152904
RecipientTypeDetails : MailUniversalSecurityGroup
PrimarySmtpAddress   : salesteam@homebiz.beer
IsDirSynced          : False
```

`MailUniversalDistributionGroup` vs `MailUniversalSecurityGroup`. Both returned by `Get-DistributionGroup` — in Exchange terminology every mail-enabled group is a "distribution group," security context or not. That inconsistency has existed for twenty years.

The demonstration that settles it

Manage mailbox delegation

The Full Access permission allows a delegate to open this mailbox and behave as the mailbox owner.

+ Add members Delete(0) 0 items announcements X ≡

User Principal Name

Manage mailbox delegation

The Full Access permission allows a delegate to open this mailbox and behave as the mailbox owner.

 Add members

 Delete(0)

1 items





☐ User Principal Name

☐  Sales Team

Same search box, same mailbox permission dialog. The distribution group returns zero results because it cannot hold a permission. The mail-enabled security group is selectable.

Group expansion in message trace

Trace Example

 Copy report text  Prepare and email extended report

 View message in Explorer  Go Hunt for this message

Sender

evan@evanireland.tech

Recipient

announcements@homebiz.beer

Received	Processed	Delivered

Status

The message was sent to the following group: announcements@homebiz.beer. Office 365 received the message and sent it to the group members.

Group: announcements@homebiz.beer

More Information

To check if a specific member of this group received the message, perform a new Message Trace query and search for the member's email address instead of the group address.

To see who is in a group, follow these steps:

1. Go to [Groups](#).
2. Double-click the group you want to review to see its members.

Message events



>	Date (UTC-07:00)	Event	Detail
>	8/23/2026, 9:35 AM	Receive	Message received...
✓	8/23/2026, 9:35 AM	Expand DL	The message was ...
The message was sent to a distribution list (DL) that was expanded to the recipients of the DL.			
>	8/23/2026, 9:35 AM	Spam	Spam confidence ...
>	8/23/2026, 9:35 AM	Drop	Reason: [{LED=25...

```
Receive → Expand DL → Drop
Reason: [{LED=250 2.1.5 RESOLVER.GRP.Expanded; distribution list expanded}]
```

The Drop is benign. 250 is an SMTP success code — the parent group recipient was retired after fan-out, by design. Expanding the row and reading the full reason is the difference between a correct diagnosis and an escalation.

The member's individual trace showed **Deliver with no Receive event** — the message was received once, at the group level, and the fan-out copies began their lives already inside transport.

Deliver with no Receive means the message arrived via group expansion, not addressed directly. If a user insists a sender emailed them personally and the trace shows Deliver-only, they were on a list.

A distribution list always requires **two searches**: the group address to confirm expansion, the member address to see what happened to their copy.

The design question

Default to separate objects. Coupled objects fuse two concerns that change on different schedules, and the failure is the **Mover** in the Joiner-Mover-Leaver lifecycle:

Stage	Coupled group behavior
Joiner	Fine, arguably convenient
Mover	The failure. User transfers departments. Someone adds them to the new list for mail. Nobody removes the old membership, because that would cut off mail during handover. They now hold file access to both departments indefinitely — entitlement creep , which audits find constantly.
Leaver	Fine

Coupled objects do not fail at grant time. They fail at *change* time, months later, silently.

Mail-enable when the coupling is the point. Group-based licensing requires a security principal; a distribution group cannot do it. A finding from the previous project — Intune enrollment failing because a department group was not nested into the licensing group — is exactly that case.

Environment constraint worth knowing

Only universal groups can be mail-enabled. A standard AGDLP model is built from **global** role groups and **domain-local** resource groups. **Neither can be mail-enabled as-is.** That is

not a flaw in AGDLP — it means "just mail-enable the department group" is not an available shortcut without changing group scope.

Two defaults to check on every new group

Edit settings

Distribution group

Sends email to all members of the list.

Group email address *

announcements

Domains

@ homebiz.beer

Communication

☒ Allow people outside of my organization to send email to this Distribution group

Joining the group

☒ Open

Anyone can join this group without owner approval.

☐ Closed

Only group owners can add members. All requests to join will be automatically declined.

☐ Owner approval

Anyone can request to join this group and owners must approve the request.

Leaving the group

☒ Open

Anyone can leave this group without group owner approval.

☐ Closed

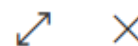
Only group owners can remove members. All requests to leave will be automatically declined.

After the group is created, you'll be able to edit settings to specify if external senders can email the group and whether or not to send copies of group conversations to members.

Setting	Risk if left at default
Allow external senders — on	Correct for a public <code>info@</code> . Wrong for an internal all-staff list — it becomes a one-address broadcast channel into the whole company for anyone on the internet. A real phishing vector.
Joining the group: Open	Low stakes on a distribution list. On a mail-enabled security group this is self-service privilege escalation.

Finding — the test was invalidated by a leftover override

The first run of this task produced a `Spam` verdict and a `Drop` on the group message, and a quarantined member copy.



Trace Example

Copy report text Prepare and email extended report

View message in Explorer Go Hunt for this message

Sender

evan@evanireland.tech

Recipient

sjohnson@homebiz.beer



Status

The message you're trying to find was identified as spam. An admin in your organization has set up a spam rule to send this kind of email to the Office 365 quarantine.

How to fix it

To release the message, follow these steps:

1. Go to the [Quarantine](#) page.
2. Click **Advanced search** to find the message.
3. Double-click the message, and then click **Release message to specified recipients**.

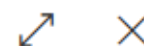
Need more help? Check out this web page's video that guides you through releasing quarantined messages: [Find and release quarantined messages as an administrator](#).

Message events



>	Date (UTC-07:00)	Event	Detail
>	8/23/2026, 9:35 AM	Spam	Spam confidence ...
✓	8/23/2026, 9:35 AM	Deliver	The message was ...

The message was successfully delivered to the folder:
DefaultFolderType:QuarantinedEmailSecured



Trace Example

- Copy report text Prepare and email extended report
- View message in Explorer Go Hunt for this message

Sender

evan@evanireland.tech

Recipient

sjohnson@homebiz.beer

Received

Processed

Delivered

Status

The message was delivered to the recipient's mailbox. Because of an Inbox rule the recipient set up, the message was delivered to the following folder:

Folder:

More Information

For help with changing your Inbox rules, see [Organize email by using Inbox rules in Outlook on the web](#).

Message events



>	Date (UTC-07:00)	Event	Detail
>	8/23/2026, 9:35 AM	Spam	Spam confidence ...
>	8/23/2026, 9:35 AM	Deliver	The message was ...
>	8/23/2026, 9:45 AM	Receive (with ext...	Message received...
✓	8/23/2026, 9:45 AM	Deliver	The message was ...

The message was successfully delivered.

The cause was not group behavior. **The Task 2 anti-spam changes had not been reverted.** The test demonstrated spam filtering acting on a message that happened to arrive via a distribution group.

Compounding it, an **allowed-sender entry** was then added to stop the interference — which meant the subsequent "is it reverted?" test would deliver successfully whether or not the revert had taken. **The verification could no longer distinguish the two states.**

(The allow list is also a genuine security anti-pattern, addressed in the findings section.)

Task 5 — Aliases and primary SMTP address

proxyAddresses

Every mail-enabled object carries the multi-valued `proxyAddresses` attribute. **The prefix case is the entire mechanism.**

```
SMTP:jsmith@contoso.beer      ← UPPERCASE = PRIMARY (reply address)
smtp:john.smith@contoso.beer  ← lowercase = alias, receives only
smtp:jsmith@contoso.onmicrosoft.com ← tenant routing address
SIP:jsmith@contoso.beer      ← Teams presence / calling identity
```

Exactly one `SMTP:` entry is permitted. "Changing someone's email address" means promoting one entry and demoting another — the old address stays as an alias so existing mail keeps arriving.

The hybrid boundary

←

×

Manage username and email

This user is synchronized with your local Active Directory. Some details can be edited only through your local Active Directory.

If the primary email is also their username, then changing the primary email will also change their current username. An alias is another email address that people can use to email John Smith.

Primary email address and username

jsmith@homebiz.beer

Aliases

Username: john.smith

Domains: homebiz.beer

Add

"This user is synchronized with your local Active Directory. Some details can be edited only through your local Active Directory."

Synced objects are mastered on-premises. The Add button is greyed out.

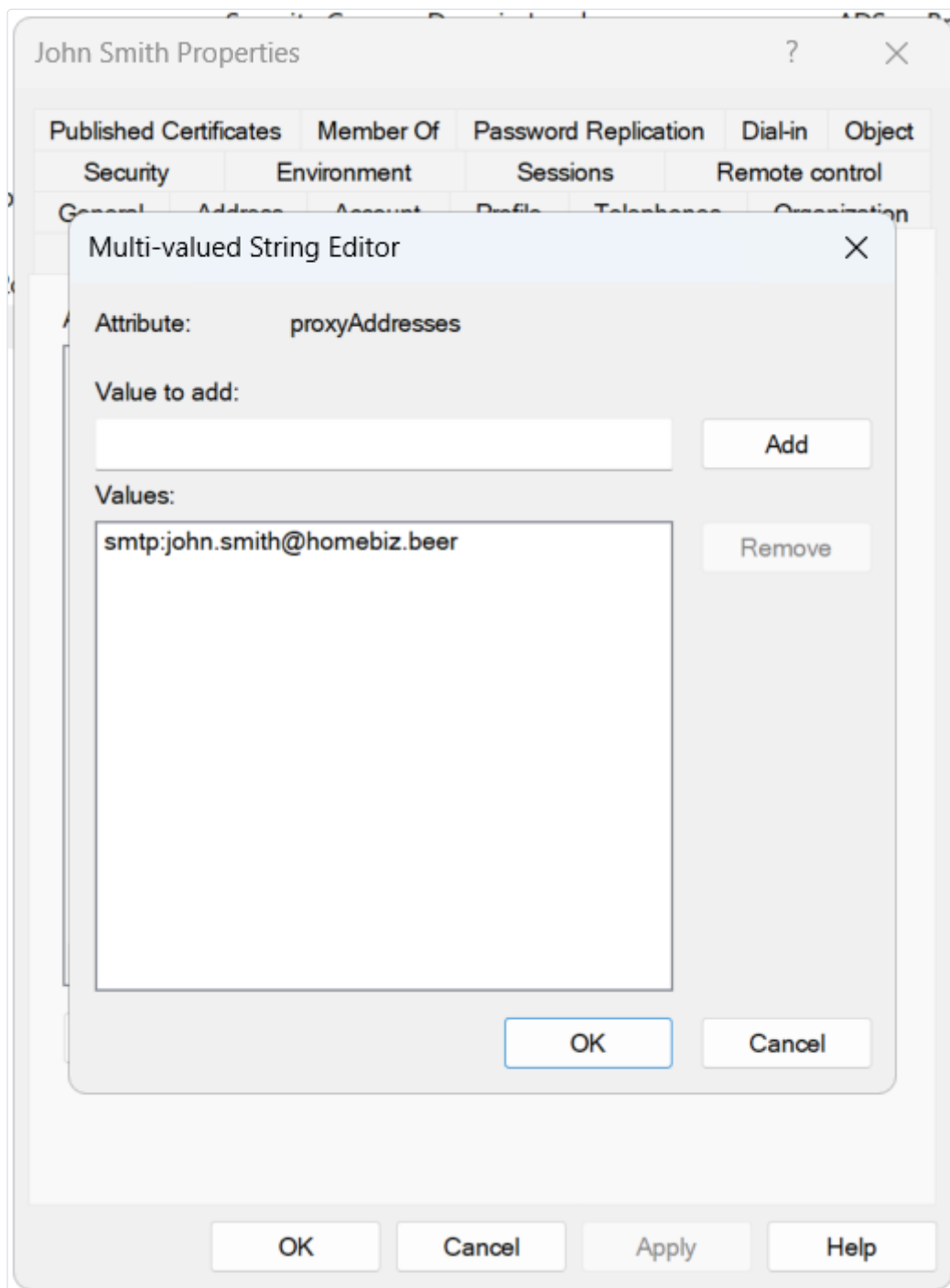
The dangerous variant of this symptom is not the greyed-out field. It is a change that appears to save and then **silently reverts on the next sync cycle**. It looks like it worked. Half an hour later it is gone.

```
Administrator: Windows PowerShell
PS C:\Users\evan> Get-Mailbox -Identity jsmith@homebiz.beer | Format-List Name,IsDirSynced,UserPrincipalName,PrimarySmtpAddress,EmailAddresses

Name                : d1ae7d41-5be8-4207-9654-ac8745446584
IsDirSynced         : True
UserPrincipalName    : jsmith@homebiz.beer
PrimarySmtpAddress   : jsmith@homebiz.beer
EmailAddresses       : {smtp:jsmith@homebizbeer.onmicrosoft.com, SIP:jsmith@homebiz.beer, SMTP:jsmith@homebiz.beer}
```

`IsDirSynced: True`. Check this before hunting for a permissions problem.

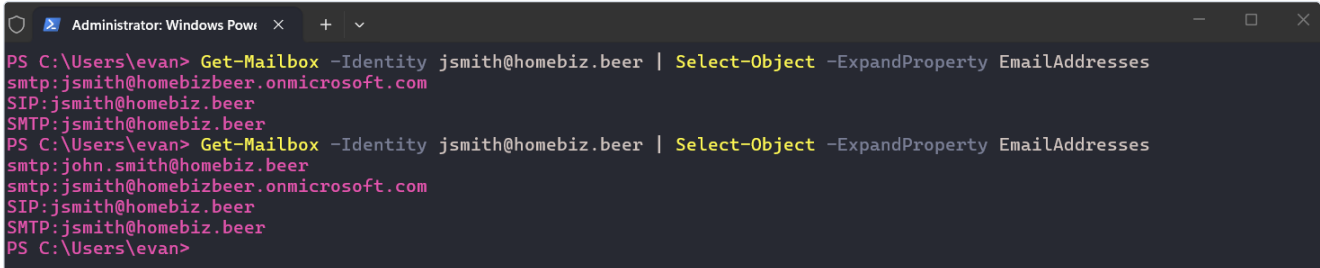
The correct procedure



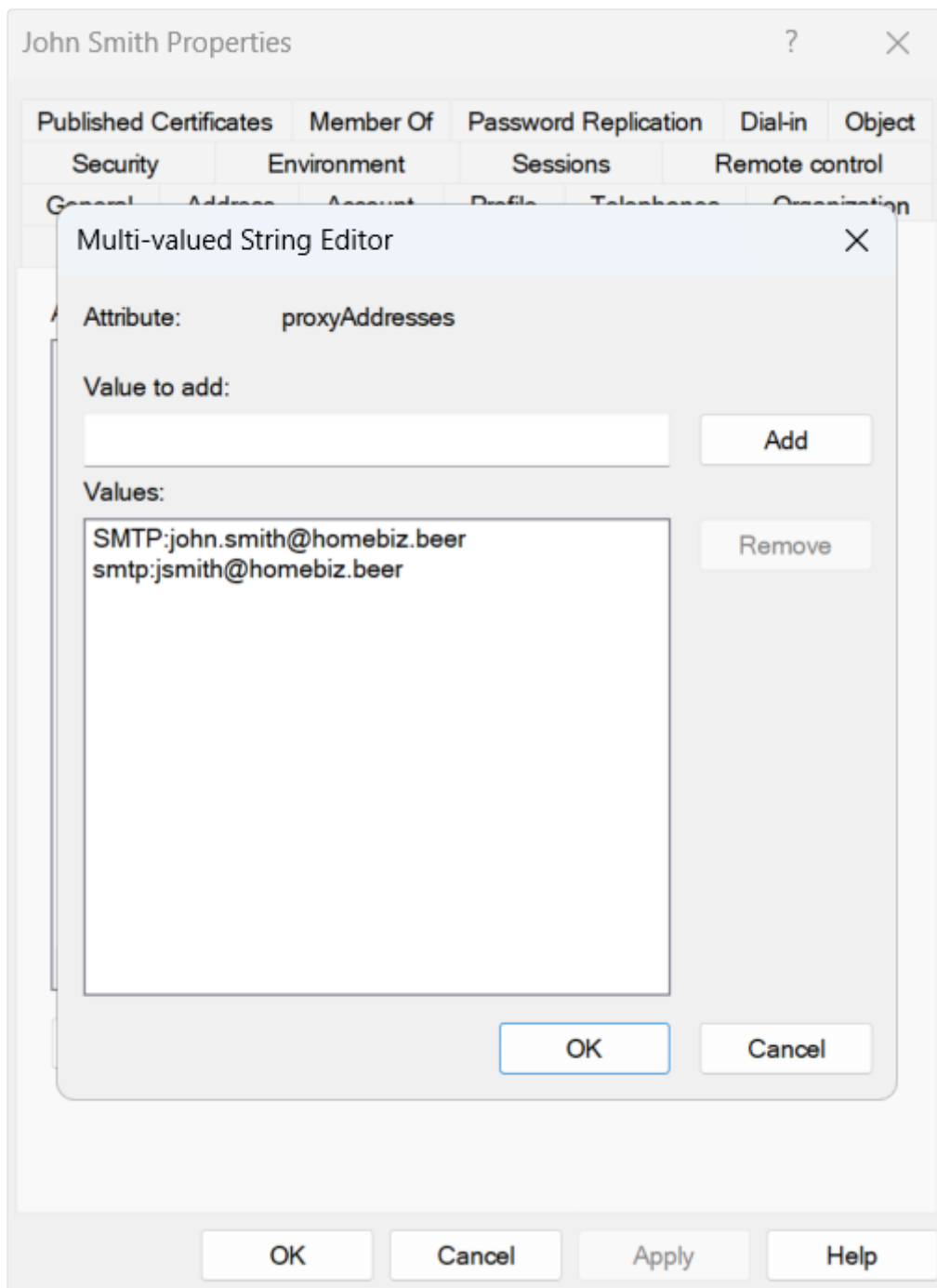
```
Set-ADUser -Identity jsmith -Add @{proxyAddresses="smtp:john.smith@contoso.beer"}
Start-ADSyncSyncCycle -PolicyType Delta
```

-Add, **never -Replace**. **-Replace** overwrites the entire multi-valued array and would remove the `onmicrosoft.com` routing address and the `SIP:` entry.

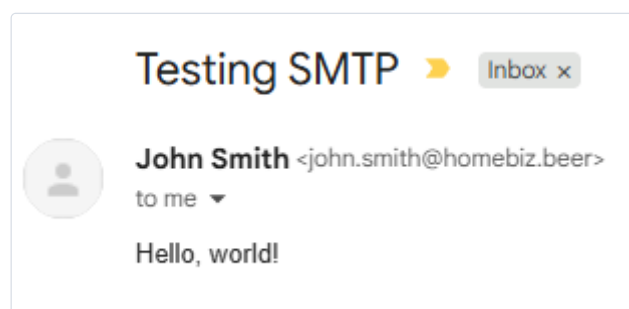
Delta, not Initial. A delta cycle processes only changed objects and is correct for attribute edits. **Initial** is a full resync, needed when a *rule or filter* changed.



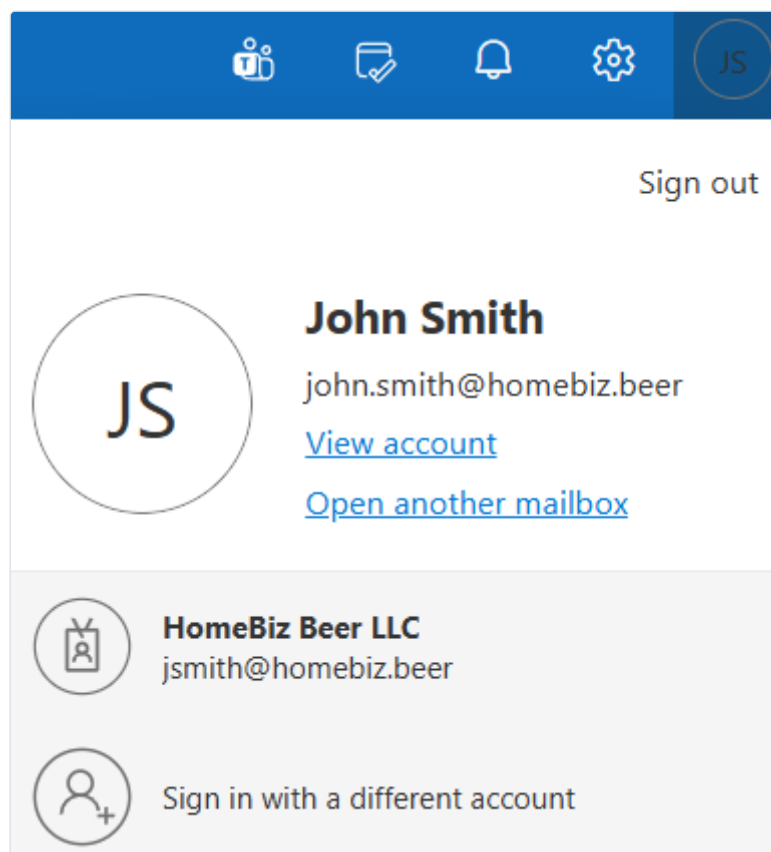
```
Administrator: Windows PowerShell
PS C:\Users\evan> Get-Mailbox -Identity jsmith@homebiz.beer | Select-Object -ExpandProperty EmailAddresses
smtp:jsmith@homebizbeer.onmicrosoft.com
SIP:jsmith@homebiz.beer
SMTP:jsmith@homebiz.beer
PS C:\Users\evan> Get-Mailbox -Identity jsmith@homebiz.beer | Select-Object -ExpandProperty EmailAddresses
smtp:john.smith@homebiz.beer
smtp:jsmith@homebizbeer.onmicrosoft.com
SIP:jsmith@homebiz.beer
SMTP:jsmith@homebiz.beer
PS C:\Users\evan>
```



Promotion: the alias goes uppercase, the old primary goes lowercase. The on-premises `mail` attribute was updated to match — a separate single-valued attribute that many line-of-business applications read instead of `proxyAddresses`. Leaving it stale is the classic cause of "the new address works everywhere except the ticketing system."



Finding — UPN is not primary SMTP



One panel, two different strings. `john.smith@contoso.beer` is the mailbox primary SMTP — the reply address. `jsmith@contoso.beer` beneath the organization name is the UPN — the sign-in identity.

Separate attributes. Entra Connect syncs both. Editing one never touches the other.

The user now emails as one string and signs in as another, which generates the ticket *"I changed my email but it says my password is wrong."* They are typing the new address at the sign-in prompt.

In production, change both and communicate that sign-in is changing. In this lab the UPN was deliberately left alone: both workstations are hybrid Entra joined and Intune enrolled against it, and a UPN change forces re-authentication across every cached credential and token on the device.

Open item — carried forward honestly. Field testing appeared to show sign-in succeeding with *both* the old UPN and the new primary SMTP. That is **not** the documented default. Microsoft's *Email as an alternate login ID* enables sign-in with `proxyAddresses`, but requires explicit enablement via Home Realm Discovery policy or staged rollout, and it is a preview feature that **explicitly does not support Entra hybrid joined devices**. The likelier explanation is a cached browser session plus OWA displaying the primary SMTP address in the account panel rather than the sign-in identifier. **Recorded as unverified pending a cold-session test.**

Task 6 — SharePoint external sharing

Three layers, most restrictive wins

Sharing

Use these settings to control sharing at the organization level in SharePoint and OneDrive. [Learn more about managing sharing settings](#)

 [Analyze my settings](#) [Show all settings](#) [Show sharing settings](#)

External sharing

Content can be shared with:

 SharePoint

 OneDrive

Most permissive

Least permissive

Anyone
Users can share files and folders using links that don't require sign-in.

New and existing guests
Guests must sign in or provide a verification code.

Existing guests
Only guests already in your organization's directory.

Only people in your organization
No external sharing allowed.

You can further restrict sharing for each individual site and OneDrive. [Learn how](#)

[More external sharing settings](#) ▾

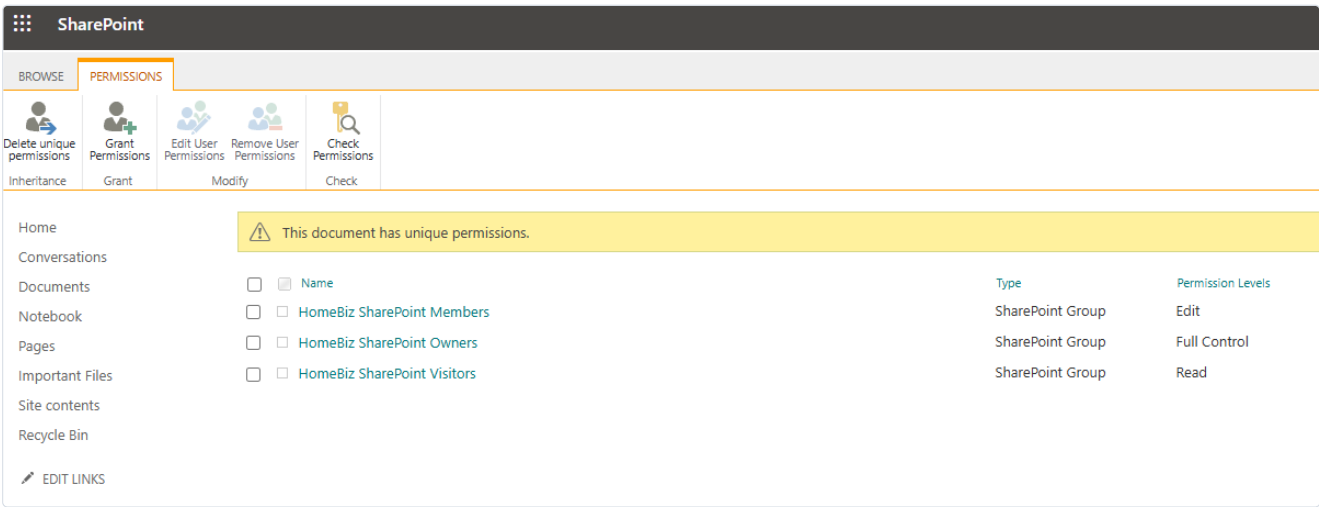
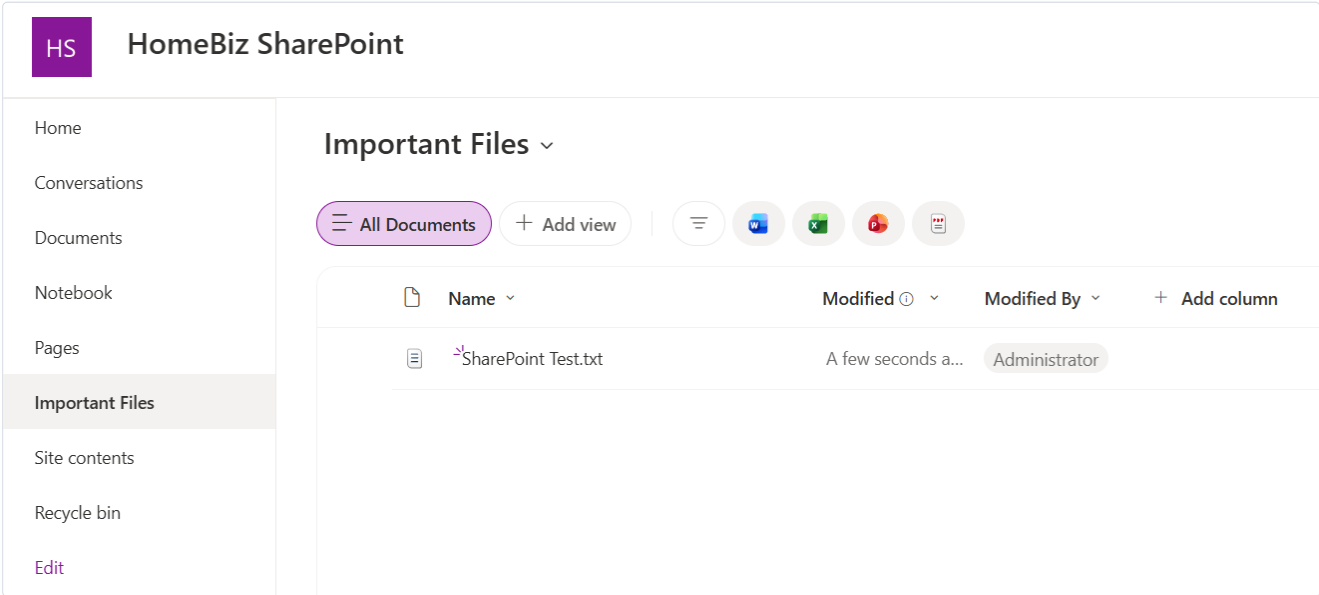
Both sliders at Anyone — the most permissive setting available. Any licensed user could generate an anonymous link to any file they could access: no sign-in, no identity, no audit trail tied to a person, and the link works for whoever it is forwarded to.

Layer	Where	Scope
Tenant	SharePoint admin center → Policies → Sharing	Ceiling for everything
Site	Active sites → site → External sharing	Cannot exceed tenant
Item	The Share dialog	Cannot exceed site

Diagnose top down. The user sees a greyed-out option at the item level; the cause is usually two layers up.

SharePoint and OneDrive are set independently — OneDrive more permissive than SharePoint is a common misconfiguration, easy to miss because the sliders sit side by side.

Breaking inheritance

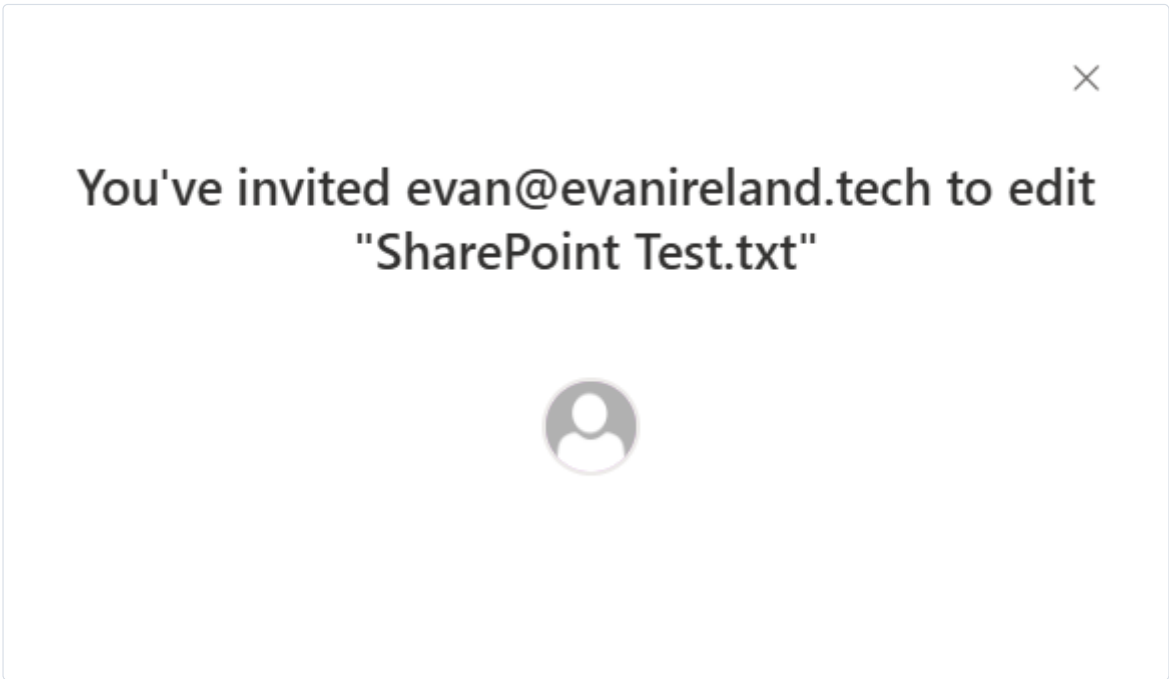


"This document has unique permissions." The list immediately afterward is an exact copy of what was inherited — breaking inheritance restricts nothing by itself, which is why people assume it failed.

Every broken-inheritance object is a permission set nobody will ever review again. It does not surface when you audit the site and it silently diverges from the parent forever. In a Copilot tenant this is precisely where surprises live — Copilot surfaces anything a user technically has access to, so permissions people forgot about become search results.

Finding — the people picker trap

The first external share produced no email. Not in the inbox, not in spam. No guest object appeared in Entra.



"You've invited Evan Ireland to edit 'SharePoint Test.txt'"

The break-glass Global Administrator account carried the display name "Evan Ireland." **The people picker resolved to it.** The file had been shared internally, with an emergency access account, and the dialog had said so.

Every piece of evidence fit:

Observation	Explanation
No email anywhere, not even spam	Nothing was ever sent externally
No guest object in Entra	No external user was invited
Confirmation named a person	It resolved to a directory object
Post-inheritance permissions showed no external principal	Confirms it

The dialog named the recipient. It was the first screenshot taken, and it was read past.

Two corrective actions:

1. **Type the full address; do not accept an autocomplete suggestion when sharing externally.**

2. **Rename the break-glass account** to something unmistakable — **BREAKGLASS - Emergency Access (DO NOT USE)** . Display name only; the UPN is the credential used in an actual lockout and must stay exactly as documented. A break-glass account named after a person is both a picker hazard and an audit-legibility problem: its sign-ins should look alarming in a log, not routine.

Display name	User principal name	User type	Is Agent	On-premises sy...	Sign-in identifiers	Company name	Creation type
Administrator	sysadmin@homebiz.beer	Member	No	No	View		
BREAKGLASS	[redacted]	Member	No	No	View		
evan	evan_evanireland.tech#EXT#@homebizbeer.onmicrosoft.com	Guest	No	No	View		Invitation
John Smith	jsmith@homebiz.beer	Member	No	Yes	View		
Sales	sales@homebiz.beer	Member	No	No	View		
Sarah Johnson	sjohnson@homebiz.beer	Member	No	Yes	View		

Finding — sharing invitations are not in message trace

Reaching for message trace here wastes your time. SharePoint sharing invitations are sent by SharePoint's own notification infrastructure from **no-reply@sharepointonline.com** . They do not originate from a mailbox in the tenant and do not appear as outbound messages in an Exchange Online trace.

Knowing which tool does not apply is as valuable as knowing which does. Someone who traces for twenty minutes and finds nothing concludes the system is broken.

Microsoft documents two causes for a sharing invitation not arriving:

1. The sending address is flagged as spam by the recipient's provider.
2. **The `Email` property on the sender's Entra user profile is empty.** The invitation is sent *on behalf of* the sharing user and pulls that identity from Entra → Contact Information → Email — **a separate field from the mailbox's primary SMTP address.**

(Same class of problem as the on-premises `mail` attribute in Task 5: the same logical concept lives in several attributes across several systems, and they do not update each other.)

Ceiling vs mechanism

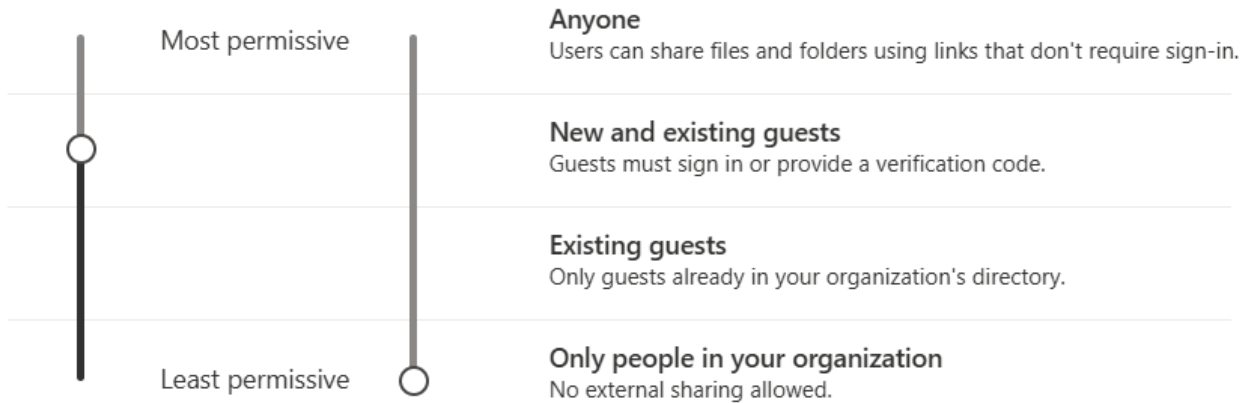
The successful external share created a guest object and required the recipient to enter a verification code.

External sharing

Content can be shared with:

 SharePoint

 OneDrive



You can further restrict sharing for each individual site and OneDrive. [Learn how](#)

Closing state: SharePoint returned to New and existing guests, OneDrive to organisation-only.

external.user_example.com#EXT#@contoso.onmicrosoft.com Guest Invitation

Original address with @ replaced by _, then the #EXT# marker, then the host tenant's initial domain. Guests live in your namespace, not theirs — which is why they appear in your people pickers and why #EXT# is the string to filter on when auditing external access.



evan@evanireland.tech

Permissions requested by:

HomeBiz Beer LLC
homebizbeer.onmicrosoft.com

By accepting, you allow this organization to:

- ✓ Receive your profile data
- ✓ Collect and log your activity
- ✓ Use your profile data and activity data

You should only accept if you trust HomeBiz Beer LLC. **HomeBiz Beer LLC has not provided links to their terms for you to review.** You can update these permissions at <https://myaccount.microsoft.com/organizations>.
[Learn more](#)

This resource is not shared by Microsoft.

Cancel

Accept

This is not "Anyone" behavior. Anonymous links require no sign-in, create no directory object, and issue no code. A named-recipient share is a **different mechanism** with a lower minimum requirement:

Mechanism	Minimum tenant setting	Creates guest?	Verification
Anonymous "Anyone" link	Anyone	No	None
Specific external person	New and existing guests	Yes	Email code or sign-in
Existing guest only	Existing guests	No	Sign-in

The tenant setting was the **ceiling**. The share only required **New and existing guests**. Had the tenant been at *Existing guests*, this exact share would have failed. Client-facing consequence: **"can we still share with clients if we turn off Anyone links?" — yes.**

Second trap in the same task: with the tenant set to Anyone, the Share dialog still did not offer an Anyone link. Which link types appear is governed by a separate **File and folder links** section further down the same settings page, and by the site-level setting. **Three places to check, not one.**

Access finding: the *Manage access* surface required the **SharePoint Administrator** role. The correct response is to grant the one role diagnosed, not a stack. The production answer to standing admin roles is **Privileged Identity Management** — eligible rather than active, elevated just-in-time. **PIM requires Entra ID P2; Business Premium includes P1.** Not available on this SKU.

Task 7 — Sensitivity labels and DLP

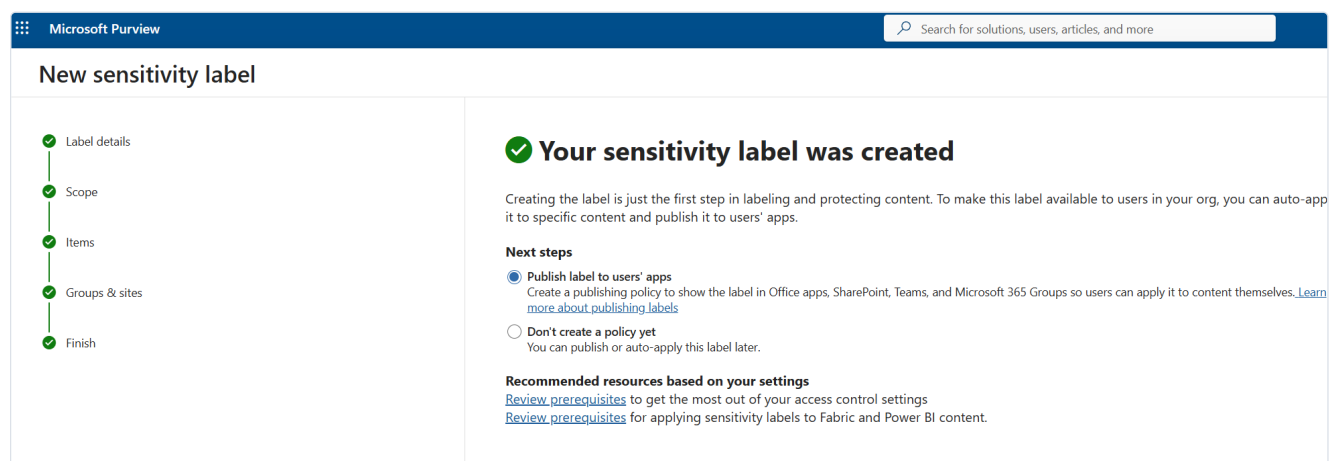
The heaviest-weighted domain on the AB-900 exam outline, and the task with the most consequential finding in the project.

Business Premium licensing boundary

Included	Requires E5 / E5 Compliance
Creating and publishing sensitivity labels	Auto-labeling — client-side and service-side
Manual labeling in Office apps	Sensitivity labels as a DLP condition
Encryption / RMS on labels, headers, footers, watermarks (AIP P1 — E3 not required)	Endpoint DLP
Scoping a label to a specific group	Insider Risk Management, Communication Compliance
DLP for Exchange Online, SharePoint, OneDrive	Custom trainable classifiers, Exact Data Match

Design consequence: build labels for manual application and build DLP on **sensitive information types**, not on labels. A label-conditioned rule on Business Premium **saves without complaint and silently never matches** — the worst possible failure mode.

Creating a label is not publishing it



Microsoft Purview

Search for solutions, users, articles, and more

New sensitivity label

- Label details
- Scope
- Items
- Groups & sites
- Finish

✔ Your sensitivity label was created

Creating the label is just the first step in labeling and protecting content. To make this label available to users in your org, you can auto-apply it to specific content and publish it to users' apps.

Next steps

- ☒ **Publish label to users' apps**
Create a publishing policy to show the label in Office apps, SharePoint, Teams, and Microsoft 365 Groups so users can apply it to content themselves. [Learn more about publishing labels](#)
- ☐ **Don't create a policy yet**
You can publish or auto-apply this label later.

Recommended resources based on your settings

- [Review prerequisites](#) to get the most out of your access control settings
- [Review prerequisites](#) for applying sensitivity labels to Fabric and Power BI content.

Two separate operations in two separate places. Create under Information Protection → Sensitivity labels; publish under **Label policies**. A label that exists but is not published appears nowhere — the single most common "*I created a label and nobody can see it*" ticket.

Deferred, honestly: the label was created and published but had not appeared in client apps at the close of the project. Microsoft documents up to 24 hours for label policy propagation.
Verification deferred rather than claimed.

DLP policy

Customize advanced DLP rules

The rules here are made up of conditions and actions that define the protection requirements for this policy. You can edit existing rules or create new ones.

[+ Create rule*](#)

1 item

Name	Status			
 Credit Card Number	 On			
Conditions Content contains any of these sensitive info types: Credit Card Number Evaluate predicate for Message or attachment				
Actions Encrypt messages with this protection setting: Encrypt Send alerts to Administrator				

Review and finish

Create the policy if these details look fine. Otherwise, adjust the settings to better meet your needs.

Basic settings

The information to protect

Edit

Custom policy

Name

Edit

Credit Card Number

Description

Edit

Create a custom policy from scratch. You will choose the type of content to protect and how you want to protect it.

Locations

Edit

Exchange email - All accounts

Mode

Edit

Run the policy in simulation mode

Rules

Edit

Credit Card Number

Field	Value
Location	Exchange email
Condition	Content contains Credit Card Number , confidence High
Action	Encrypt; send alerts to Administrator
Mode	Simulation

 Policy tip: 'This item contains 1 or more credit card numbers and is being shared with people outside your organization. Consider removing the numbers or make sure it's OK that everyone you're sharing with can view them.' [Show details](#)

 Send

▼

From: sales@homebiz.beer ▼

 ▼
 


To

evan@evanireland.tech

×

Bcc

Cc

Credit Card Number Test

Draft saved at 6:41 PM

Hey Evan,

Here's my credit card number: 4111 1111 1111 1111
Exp: 01/31 CSC: 123

Best,

 Systems Administrator

The policy tip fires in the client before send.

Finding — simulation mode produces the full appearance of enforcement




Credit Card Number Test

 Copy report text
  Prepare and email extended report

 View message in Explorer
  Go Hunt for this message

Sender	Recipient
sysadmin@homebiz.beer	evan@evanireland.tech

Received

Processed

Sent

Status

Office 365 successfully sent the message to the following external address:

External address: evan@evanireland.tech

More Information

If the recipient can't find the message in their Inbox folder, one of several things could have happened:

- The message may have been deleted or moved to another folder (for example, the Junk Email folder).
- The recipient might have created a forwarding rule that forwarded the message to a different email address.
- There might be something wrong with the recipient's email system.

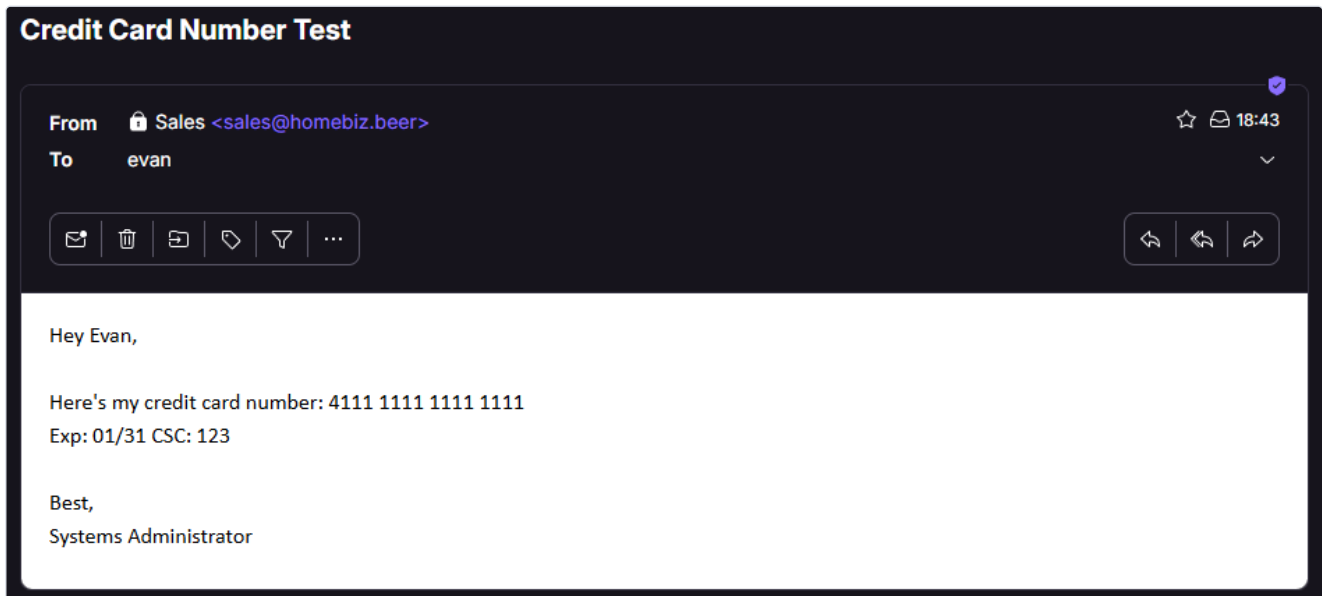
Until the problem is resolved, ask the sender to try contacting the recipient another way. For instance, send the message to a different email address.

Message events

>	Date (UTC-07:00)	Event	Detail
>	8/23/2026, 6:43 PM	Receive	Message received by...
>	8/23/2026, 6:43 PM	Submit	The message was su...
>	8/23/2026, 6:43 PM	DLP rule	DLP rule: ", ID: ('17D...
>	8/23/2026, 6:43 PM	DLP rule	DLP rule: ", ID: ('B97...
>	8/23/2026, 6:43 PM	DLP rule	DLP rule: ", ID: ('B97...
>	8/23/2026, 6:43 PM	TRANSFER	
>	8/23/2026, 6:43 PM	Send external	Message sent to mai...

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The trace showed **three DLP rule evaluations**. The client showed a policy tip. The sender received a notification: *"You shared content that contains 1 or more credit card numbers with people outside your organization."*



And the card number arrived at the external recipient in plain text.

The policy was then switched to enforcement and the identical test repeated thirty-three minutes later:

Credit Card Number

 Copy report text  Prepare and email extended report

 View message in Explorer  Go Hunt for this message

Sender

sysadmin@homebiz.beer

Recipient

evan@evanireland.tech

Received

Processed

Sent

Status

Office 365 successfully sent the message to the following external address:

External address: evan@evanireland.tech

More Information

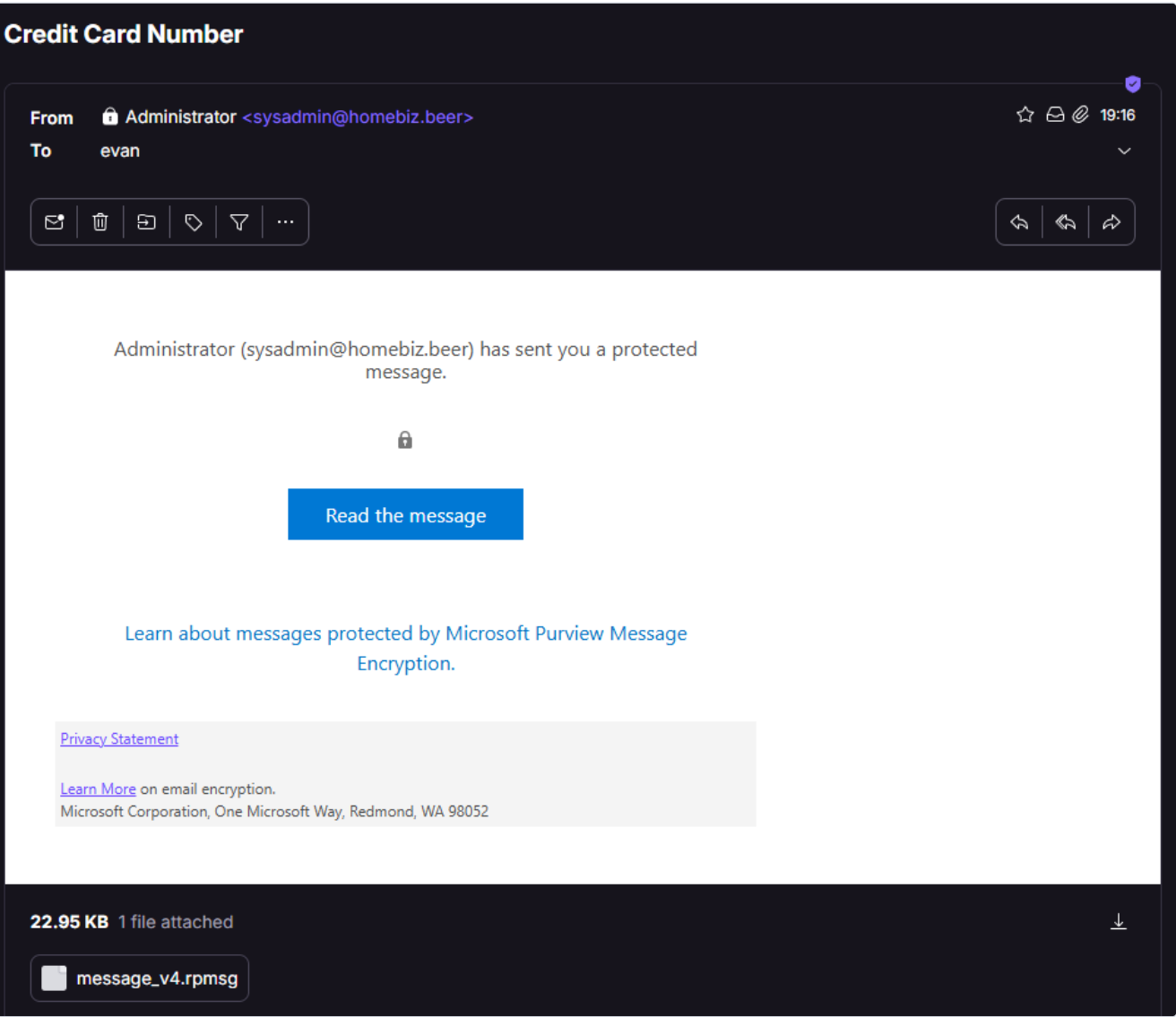
If the recipient can't find the message in their Inbox folder, one of several things could have happened:

- The message may have been deleted or moved to another folder (for example, the Junk Email folder).
- The recipient might have created a forwarding rule that forwarded the message to a different email address.
- There might be something wrong with the recipient's email system.

Until the problem is resolved, ask the sender to try contacting the recipient another way. For instance, send the message to a different email address.

Message events

▼	Date (UTC-07:00)	Event	Detail
▼	8/23/2026, 7:16 PM	Receive	Message received by: CH3P220MB2050.NAMP220.PROD.OUTLOOK.COM
Message received by: CH3P220MB2050.NAMP220.PROD.OUTLOOK.COM			
▼	8/23/2026, 7:16 PM	Submit	The message was submitted.
The message was submitted.			
▼	8/23/2026, 7:16 PM	DLP rule	DLP rule: ", ID: ('B975FCAF-5B2C-4777-9D6D-22B8B821CD70'), DLP policy: ", ID: (0000...
DLP rule: ", ID: ('B975FCAF-5B2C-4777-9D6D-22B8B821CD70'), DLP policy: ", ID: (00000000-0000-0000-0000-000000000000).			
▼	8/23/2026, 7:16 PM	DLP rule	DLP rule: ", ID: ('B975FCAF-5B2C-4777-9D6D-22B8B821CD70'), DLP policy: ", ID: (0000...
DLP rule: ", ID: ('B975FCAF-5B2C-4777-9D6D-22B8B821CD70'), DLP policy: ", ID: (00000000-0000-0000-0000-000000000000).			
▼	8/23/2026, 7:16 PM	DLP rule	DLP rule: ", ID: ('17D0F803-1534-4751-A21F-B9B8C5BFA46E'), DLP policy: ", ID: (00000...
DLP rule: ", ID: ('17D0F803-1534-4751-A21F-B9B8C5BFA46E'), DLP policy: ", ID: (00000000-0000-0000-0000-000000000000).			
▼	8/23/2026, 7:16 PM	Send external	Message sent to mail.protonmail.ch at 185.70.42.128
Message sent to mail.protonmail.ch at 185.70.42.128			



	Simulation, 6:43 PM	Enforcement, 7:16 PM
Message trace events	Receive → Submit → DLP rule ×3 → Send external	Receive → Submit → DLP rule ×3 → Send external
Policy tip in client	Yes	Yes
User notification	Yes	Yes
What the recipient received	Plaintext card number	Encrypted .rpmsg wrapper

The message trace is effectively identical in both modes. Neither the trace nor the policy tip can distinguish a simulated match from an enforced one.

The only reliable verification that DLP is enforcing is confirming the action occurred at the far end — the message was blocked, encrypted, or the recipient received something different from what was sent. **Never accept a policy tip as evidence of enforcement.**

An administrator who deployed this and walked away would believe the tenant was protected. It was not.

Finding — Encrypt is not Block

Note that the enforcement trace **still ends in Send external**. Encrypt wraps and delivers; the recipient authenticates to open it. The data still leaves — protected in transit and at rest, not withheld.

If the requirement is "this must not leave," the action is Block. Clients conflate these constantly and the distinction matters in a compliance conversation.

Supporting observations

Confidence level is the noise dial. The Credit Card Number classifier requires a number passing the Luhn checksum **plus corroborative evidence** — an expiration date, a security code, **or a keyword such as "credit card."** Any one suffices, which is why a second test omitting the expiry and CSC still matched at High confidence.

 Policy tip: 'This item contains 1 or more credit card numbers and is being shared with people outside your organization. Consider removing the numbers or make sure it's OK that everyone you're sharing with can view them.' [Show details](#)

 Send

From: sysadmin@homebiz.beer

To

evan@evanireland.tech

Bcc

Cc

Credit Card Number

Draft saved at 7:15 PM

Hey Evan,

Sorry I don't know if my last email went through: Here's the credit card number you asked for:

4111 1111 1111 1111

Let me know if that works out!

 Systems Administrator

When a client says "DLP isn't catching things," confidence threshold is the first setting to check — and it is the same setting when they say "DLP blocks everything."

Reporting is batched, not real-time. Policy tips and trace events are inline and instant; alerts, reports, and Activity Explorer lag by hours. **An empty report five minutes after a test proves nothing.** Simulation results appear only in the policy's own simulation view, not in the global alerts surface.

Two notification paths, two audiences. The user notification ("*You shared content...*") is a different mechanism from the `Send alerts to Administrator` rule action. Receiving one does not mean the other fired.

Check what is already running. The trace showed more DLP rule evaluations than policies created, with distinct rule IDs — Microsoft-provisioned default policies ship enabled in newer tenants. "*What is already running here that I did not put there?*" is the first question in any inherited environment.

Cross-cutting findings

The three most valuable findings were not specific to any product. Each appeared in a different service, and together they form a doctrine.

1. An override makes your diagnostics lie

Encountered three times, in three unrelated products, in a day and a half.

Override	What it broke
Wildcard * CNAME in the DNS zone	Every "does this record exist?" query returned a valid answer. Five missing records all appeared present.
Allowed-sender entry in an anti-spam policy	Every "is filtering working again?" test delivered successfully, whether or not the revert had taken.
DLP policy in simulation mode	Policy tips fired, message trace logged rule events, the user got a notification — and the data left unprotected.

A configuration override does not only change behavior. It changes what your tests *report*, and it does so silently. The test still runs, still returns a result, and the result still looks like an answer.

Rules:

1. **Remove overrides before testing, not after.** A revert verified while an override is active is not verified.
2. **Verify at the far end.** DNS answering is not mail flowing. A policy tip is not enforcement. Check what the recipient actually received.
3. **When a result matches your expectation, ask what the test could not have told you.** Agreement is not confirmation.
4. **In an inherited environment, inventory the overrides before trusting any diagnostic.** Wildcard DNS records, allow lists, disabled policies, and simulation-mode policies are all invisible until you look for them specifically.

2. Read the screen before you diagnose

Incident	What the screen already said
DNS delegation failing for 45+ minutes	"DNSSEC is not supported by Microsoft 365 DNS hosting" — a banner in the admin center
SharePoint share "never arrived"	The confirmation dialog named the actual recipient — a different, internal account
Unexplained <code>Drop</code> in a message trace	The detail read <code>LED=250 2.1.5 RESOLVER.GRP.Expanded</code> — <code>250</code> is a success code

Before forming a theory, re-read every banner, confirmation, and error detail already on screen. Expand the truncated ones. Diagnostic effort spent before this step is usually wasted.

3. Change one thing at a time

Two failures in this project could not be attributed cleanly because two variables moved together — a Task 2 revert paired with an allow-list addition, and a missing SharePoint role granted alongside two others that were never tested against a symptom.

When two things change at once you have lost causality, and in a client environment that is the difference between a root cause and a guess you will repeat.

Corrections to prior understanding

Previously believed	Correct
GTUBE is a universal spam-filter test string	Not documented as supported by EOP; does not reliably trip it
Business Premium sensitivity labels cannot do encryption	They can — Business Premium includes AIP P1, covering RMS encryption and visual markings. E3 is not required
Sensitivity labels can be used as DLP conditions on any SKU	Business Premium's DLP engine does not evaluate labels. Also a condition , not an action
A policy tip proves DLP is protecting data	Policy tips and trace DLP events fire identically in simulation mode
DKIM selector CNAME targets follow a predictable pattern	Format changed May 2025; includes a dynamically assigned partition character. Must be read from the portal
Publishing DKIM CNAMEs and enabling signing are always two distinct manual steps	Depends on the path used to obtain the values. Never assume the toggle state — check it
Changing primary SMTP means the old address can no longer be used to sign in	The old address remains the UPN and still works. Whether the <i>new</i> address also works depends on a feature that is off by default — open item

Deferred and unverified

Recorded rather than glossed over.

Item	Status
Sensitivity label visibility in client apps	Created and published; had not propagated at close. Microsoft documents up to 24 hours. Verification deferred.
Sign-in with a non-UPN proxy address	Observed but not cleanly tested — likely a cached session plus OWA display behavior. Requires a cold-session test. Unverified.
Anonymous "Anyone" link end-to-end test	Blocked by the File and folder links setting; not completed before the tenant was returned to <i>New and existing guests</i> .
Admin alert delivery from the DLP rule	User notification confirmed; admin alert not separately confirmed against the batched reporting pipeline.

Closing state

Item	Final state
Public DNS	Full record set published. SPF, DKIM, DMARC all passing at an external receiver
DMARC	<code>p=none</code> — monitoring only, not enforcement. Ladder to <code>quarantine</code> and <code>reject</code> documented
Anti-spam policy	Reverted to defaults. Allowed-sender entry removed
Shared mailbox	<code>sales@</code> with Full Access, Send As, Send on Behalf; Sent Items copy enabled
Groups	One distribution group, one mail-enabled security group, both cloud-managed
Mailbox addresses	Alias added and promoted on-premises; <code>mail</code> attribute aligned; UPN deliberately unchanged
SharePoint sharing	Returned to New and existing guests . One guest object retained as an artifact
Purview	One label created and published; one DLP policy enforcing on Exchange email
DNSSEC	Off. To be re-enabled as a separate, deliberate change with independent verification

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